
Computer Networks

Complete Study Notes — Modules 1 to 7

Syllabus Coverage

- Module 1** Introduction, Topologies, OSI & TCP/IP Models
- Module 2** Physical Layer: Transmission Media, Multiplexing, Switching
- Module 3** Data Link Layer: ARQ, HDLC, MAC Protocols, IEEE 802
- Module 4** Network Layer: IP, Routing Algorithms, NAT, ICMP
- Module 5** Transport Layer: UDP, TCP, Congestion Control
- Module 6** Application Layer: DNS, HTTP, FTP, SMTP, SNMP
- Module 7** Emerging Concepts: Cloud, IoT, MQTT, CoAP, SDN

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Note

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Module 1 — Introduction to Computer Networks

Introduction, Topologies, OSI & TCP/IP Reference Models

6 Hours · 6 Marks

1 Introduction to Computer Networking

1.1 Definition of a Computer Network

Definition: Computer Network

A **computer network** is a collection of two or more computing devices (nodes) interconnected by communication links (wired or wireless) that share resources, exchange data, and communicate with one another under a defined set of rules called *protocols*.

More formally, a network can be defined as:

$$\mathcal{N} = (V, E, P)$$

where V is the set of *nodes* (hosts, routers, switches), E is the set of communication *links*, and P is the set of governing *protocols*.

1.2 Importance and Goals of Networking

Key Concept: Why Networks Matter

Networks are the backbone of modern information technology, enabling communication, resource sharing, and distributed computing at global scale.

The primary goals and benefits of computer networking are:

1. Resource Sharing

Hardware (printers, storage, GPUs) and software (applications, databases) can be shared among multiple users, reducing cost and redundancy.

2. Data Communication

Networks facilitate rapid transmission of data, voice, and video across large distances (e.g., email, VoIP, video conferencing).

3. High Reliability

By providing multiple paths between source and destination, networks offer fault tolerance – if one path fails, traffic is rerouted automatically.

4. Scalability

Additional nodes can be added to a network without redesigning the entire infrastructure.

5. Cost Reduction

Sharing expensive resources and centralizing management lowers the total cost of ownership.

6. Centralized Management

Network administrators can monitor, configure, and troubleshoot devices from a central location.

7. Enabling Distributed Computing

Tasks can be split across multiple machines (e.g., cloud computing, grid computing), enhancing performance.

2 Components of a Computer Network

A computer network is built from three broad categories of components: **hardware**, **software**, and **protocols**.

2.1 Hardware Components

2.1.1 End Devices (Hosts)

Definition: End Device / Host

An **end device** is any device that originates or consumes data in the network. Examples include computers, smartphones, servers, printers, and IoT sensors.

2.1.2 Network Interface Card (NIC)

The NIC provides the physical interface between a host and the network medium. It has a globally unique **MAC address** (48-bit) burned into its firmware.

2.1.3 Transmission Media

Table 2.1: Types of Transmission Media

Medium	Description	Typical Speed
Twisted Pair (UTP/STP)	Pairs of insulated copper wires twisted together to reduce EMI	10 Mbps – 10 Gbps
Coaxial Cable	Central copper conductor with shielding layers	10 – 500 Mbps
Optical Fiber	Pulses of light through glass/plastic core	100 Mbps – 100+ Tbps
Radio Waves (Wi-Fi)	Electromagnetic waves through free space	Up to several Gbps
Microwave / Satellite	Line-of-sight or orbital relay	Variable

2.1.4 Network Devices

Table 2.2: Key Network Devices and Their OSI Layers

Device	OSI Layer	Function
Hub	Layer 1	Broadcasts incoming signals to all ports (dumb repeater)
Switch	Layer 2	Forwards frames based on MAC address table; creates collision domains per port
Router	Layer 3	Routes packets between different networks using IP addresses
Bridge	Layer 2	Connects two LAN segments; filters traffic by MAC
Gateway	Layer 4–7	Protocol translation between different network architectures
Repeater	Layer 1	Amplifies/regenerates signals to extend cable distance
Modem	Layer 1	Modulates/demodulates signals for transmission over analog lines
Access Point	Layer 2	Wireless bridge connecting Wi-Fi clients to a wired network

2.2 Software Components

- **Network Operating System (NOS):** e.g., Cisco IOS, Windows Server
- **Protocol Software:** TCP/IP stack, DNS resolver, DHCP client
- **Network Management Software:** SNMP managers, monitoring dashboards
- **Security Software:** Firewalls, IDS/IPS, VPN clients

2.3 Protocols

Definition: Protocol

A **protocol** is a set of rules that governs how data is formatted, transmitted, received, and acknowledged between network entities. Protocols define syntax, semantics, and timing.

Examples: TCP, UDP, IP, HTTP, HTTPS, FTP, SMTP, DNS, DHCP, ARP, ICMP.

3 Network Types: LAN, MAN, and WAN

Networks are classified by their **geographical span**:

Table 3.1: Comparison of LAN, MAN, and WAN

Property	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Geographical Span	Room / Building / Campus (up to ~1 km)	City / Town (1 – 100 km)	Country / Continent / Globe
Ownership	Single organization	Single or multiple orgs / ISP	Multiple organizations, carriers
Data Rate	100 Mbps – 10 Gbps	10 Mbps – 1 Gbps	56 Kbps – 100 Gbps (varies)
Propagation Delay	Very low	Moderate	High
Error Rate	Very low	Low	Higher
Typical Technologies	Ethernet, (802.11), Wi-Fi	Metro Ethernet, WiMAX, DQDB	MPLS, Frame Relay, ATM, SONET, Internet
Examples	Office network, home Wi-Fi	City cable TV network	Internet, corporate WAN

3.1 Local Area Network (LAN)

Key Concept: LAN

A **LAN** connects devices within a limited geographic area (office, school, home). It is characterized by high data rates, low error rates, and single-organization ownership. Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11) are the dominant LAN technologies.

Key features:

- Speeds: typically 100 Mbps, 1 Gbps, 10 Gbps on modern infrastructure
- Collision domain management using switches
- Broadcast domain controlled by routers / VLANs
- Inexpensive to set up and maintain

3.2 Metropolitan Area Network (MAN)

Key Concept: MAN

A **MAN** spans a city or large campus. It is larger than a LAN but smaller than a WAN. Cable TV networks and city-wide broadband are classic examples.

Technologies:

- **DQDB** (Distributed Queue Dual Bus) – IEEE 802.6
- **WiMAX** (IEEE 802.16) – wireless MAN
- Metro Ethernet – carrier-grade Ethernet services

3.3 Wide Area Network (WAN)

Key Concept: WAN

A **WAN** spans countries or continents. It uses public or leased telecommunication infrastructure. The Internet is the largest WAN.

Technologies:

- PSTN, leased lines (T1/E1), ISDN
- Frame Relay, ATM
- SONET/SDH (optical backbone)
- MPLS (Multi-Protocol Label Switching)
- Broadband (DSL, cable, fiber to the home)

4 Network Topologies

Definition: Network Topology

Network topology refers to the arrangement of nodes and links in a network. It can be described as a *physical topology* (actual layout of cables/devices) or a *logical topology* (how data flows).

4.1 Bus Topology

Key Concept: Bus Topology

All devices are connected to a single shared communication line (the *bus*). Data sent by any node travels in both directions along the bus and is received by all other nodes.

Operation:

1. A node places data on the bus with the destination address.
2. All nodes receive the signal; only the intended recipient processes it.
3. Terminators at each end absorb signals to prevent reflection.

Table 4.1: Bus Topology – Advantages and Disadvantages

Advantages	Disadvantages
Simple and inexpensive to install	Single point of failure (bus cable)
Easy to extend by adding nodes	Performance degrades with more nodes
Less cabling required than star	Difficult to isolate faults
Suitable for small networks	Collision-prone; uses CSMA/CD

Note

Classic Ethernet (10BASE2, 10BASE5) used bus topology. Modern Ethernet is physically star but logically bus.

4.2 Star Topology

Key Concept: Star Topology

All devices connect to a **central hub or switch**. Every communication between nodes passes through this central device. This is the most widely used topology in modern LANs.

Operation:

- With a **hub**: incoming frames are broadcast to all ports.
- With a **switch**: frames are forwarded only to the destination port (unicast), reducing congestion.

Table 4.2: Star Topology – Advantages and Disadvantages

Advantages	Disadvantages
Easy to add/remove nodes	Central device is a single point of failure
Fault isolation is straightforward	More cabling than bus
High performance with switches	Cost of central switch/hub
Widely supported; scalable	Performance limited by central device capacity

4.3 Ring Topology

Key Concept: Ring Topology

Devices are connected in a closed loop (ring). Data travels in one direction (unidirectional) or both directions (bidirectional / dual ring) around the ring. Token Ring (IEEE 802.5) is the classic example.

Token Passing Mechanism:

1. A special control frame called a *token* circulates around the ring.
2. A node wishing to transmit must *capture* the token.
3. After transmission, the token is released for the next node.
4. This eliminates collisions (deterministic access).

Table 4.3: Ring Topology – Advantages and Disadvantages

Advantages	Disadvantages
No collisions (token-controlled)	Failure of one node can disrupt entire ring
Predictable / deterministic performance	Difficult to add/remove nodes
Equal access for all nodes	Slower than star with switches
FDDI dual-ring provides redundancy	More complex configuration

Note

FDDI (Fiber Distributed Data Interface) used a dual-ring topology at 100 Mbps over fiber, providing fault tolerance: if one ring fails, the network wraps around the break using the secondary ring.

4.4 Mesh Topology

Key Concept: Mesh Topology

Every node has a dedicated point-to-point link to *every other node*. This provides maximum redundancy and reliability. Commonly used in the backbone of WANs and the Internet.

Mathematical Properties:

For n nodes in a *full mesh*:

$$\text{Number of links} = \frac{n(n-1)}{2}$$

$$\text{I/O ports per node} = n - 1$$

Example 4.1. For $n = 5$ nodes:

$$\text{Links} = \frac{5 \times 4}{2} = 10, \quad \text{Ports per node} = 4$$

Table 4.4: Mesh Topology – Advantages and Disadvantages

Advantages	Disadvantages
Highest fault tolerance	Very expensive (cabling and ports)
Dedicated links ensure privacy	Complex installation and configuration
Simultaneous multiple transmissions	Does not scale well (quadratic cost)
Multiple paths for load balancing	High maintenance overhead

A **partial mesh** reduces cost by connecting only critical nodes with dedicated links and using shared links for others.

4.5 Topology Comparison Summary

Table 4.5: Summary Comparison of Network Topologies

Property	Bus	Star	Ring	Mesh
Cable needed	Low	Medium	Medium	Very High
Installation	Easy	Easy	Moderate	Complex
Fault Tolerance	Low	Medium	Low–Medium	Very High
Fault Isolation	Hard	Easy	Hard	Easy
Performance	Degrades	High (switch)	Moderate	High
Scalability	Low	High	Low	Very Low
Cost	Low	Medium	Medium	Very High
Technology Example	10BASE2	Fast Ether-net	Token Ring, FDDI	Internet core, SONET

Exam Tip

For MAKAUT exams, be prepared to:

- Draw neat labeled diagrams of each topology.
- State at least 3 advantages and 3 disadvantages of each.
- Apply the mesh formula $n(n-1)/2$ to numerical problems.
- Distinguish *physical* from *logical* topology.

5 Reference Models

Key Concept: Why Reference Models?

A **reference model** provides a conceptual framework that divides network communication into well-defined layers, each with specific responsibilities. This promotes interoperability, modularity, and vendor-independent standards.

The two dominant reference models are:

1. **OSI Model** – Open Systems Interconnection (ISO, 1984)
2. **TCP/IP Model** – ARPA/DOD model, basis of the Internet

5.1 The OSI Reference Model

Definition: OSI Model

The **OSI (Open Systems Interconnection) model** is a 7-layer conceptual framework developed by the International Organization for Standardization (ISO) in 1984 to standardize network communication between different systems regardless of their underlying architecture.

5.1.1 Memory Aid

Mnemonic (Top to Bottom): “*All People Seem To Need Data Processing*”
Application – Presentation – Session – Transport – Network – Data Link – Physical

5.1.2 Layer-by-Layer Breakdown

Layer 7 – Application Layer

Definition: Application Layer (L7)

The topmost layer; provides the interface between the network and end-user applications. It provides services such as email, file transfer, remote login, and the web directly to user programs.

- **Functions:** Network virtual terminal, file transfer, email, directory services, resource sharing.
- **Key Protocols:** HTTP/HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP, SSH, Telnet, SNMP.
- **PDU:** Message / Data.

Layer 6 – Presentation Layer

Definition: Presentation Layer (L6)

Responsible for data **translation, encryption/decryption, and compression**. It ensures that data sent by one system can be understood by the application layer of another system.

- **Functions:** Character encoding (ASCII, EBCDIC, Unicode), data compression (zip, JPEG), encryption (SSL/TLS at this layer conceptually).
- **Formats:** JPEG, MPEG, GIF, ASCII, EBCDIC.
- **PDU:** Data.

Layer 5 – Session Layer

Definition: Session Layer (L5)

Manages **sessions** (connections) between applications. It establishes, maintains, synchronizes, and terminates dialogues between communicating systems.

- **Functions:** Session establishment/teardown, checkpointing (synchronization points for recovery), dialog control (half-duplex vs. full-duplex), token management.
- **Protocols:** NetBIOS, RPC, PPTP, SIP.
- **PDU:** Data.

Layer 4 – Transport Layer

Definition: Transport Layer (L4)

Provides **end-to-end** communication services for applications. It segments data from the session layer, reassembles segments at the destination, and provides error control and flow control.

- **Functions:** Segmentation and reassembly, connection management (connection-oriented TCP vs. connectionless UDP), error detection and correction, flow control (sliding window), multiplexing via port numbers.
- **Protocols:** TCP (reliable), UDP (unreliable/fast).
- **PDU:** Segment (TCP) / Datagram (UDP).

Note

TCP vs UDP in brief:

TCP – connection-oriented, reliable, ordered delivery, flow and congestion control. Uses 3-way handshake (SYN, SYN-ACK, ACK).

UDP – connectionless, unreliable, no ordering guarantees, low overhead. Suitable for streaming, DNS, VoIP.

Layer 3 – Network Layer

Definition: Network Layer (L3)

Responsible for **logical addressing** (IP addresses) and **routing** packets from source to destination across multiple networks (*internetworking*).

- **Functions:** Logical addressing, routing (path determination), packet forwarding, fragmentation and reassembly, congestion control at the network level.
- **Protocols:** IP (IPv4, IPv6), ICMP, IGMP, ARP (conceptually sits between L2 and L3), OSPF, RIP, BGP.
- **Devices:** Router.
- **PDU:** Packet.

Layer 2 – Data Link Layer

Definition: Data Link Layer (L2)

Provides **node-to-node** (hop-to-hop) reliable transfer of data frames over a single physical link. It handles physical addressing (MAC), error detection, and access to the shared medium.

- **Functions:** Framing, physical addressing (MAC), error detection (CRC), flow control, medium access control (CSMA/CD, CSMA/CA, token passing).
- **Sub-layers:**

- **LLC** (Logical Link Control) – IEEE 802.2: flow/error control.
- **MAC** (Medium Access Control) – IEEE 802.3: access to medium.
- **Protocols:** Ethernet, PPP, HDLC, Wi-Fi (802.11), Frame Relay.
- **Devices:** Switch, Bridge.
- **PDU:** Frame.

Layer 1 – Physical Layer

Definition: Physical Layer (L1)

The lowest layer; concerned with the **physical transmission** of raw bits over a communication channel. It defines the electrical, mechanical, and functional specifications of the physical medium.

- **Functions:** Bit transmission/reception, encoding (NRZ, Manchester, 4B/5B), signal type (digital vs. analog), data rate, physical connectors and cable specs, synchronization of bits.
- **Standards:** RS-232, V.35, IEEE 802.3 (physical specs), Ethernet (100BASE-TX), DSL.
- **Devices:** Hub, Repeater, cables.
- **PDU:** Bit.

5.1.3 OSI Layer Summary Table

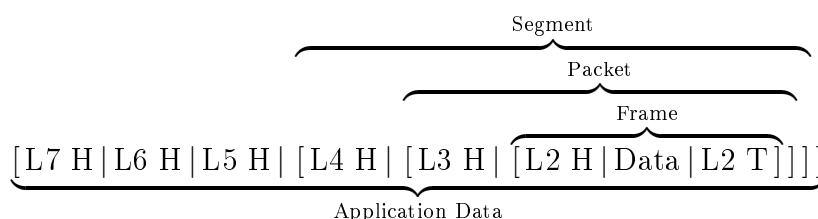
Table 5.1: OSI Reference Model – Complete Summary

No.	Layer	PDU	Key Function	Devices
7	Application	Data/Message	User interface, application services	End systems
6	Presentation	Data	Translation, encryption, compression	End systems
5	Session	Data	Session management, synchronization	End systems
4	Transport	Segment/Datagram	End-to-end delivery, error/flow control	End systems
3	Network	Packet	Logical addressing, routing	Router
2	Data Link	Frame	Physical addressing, error detection	Switch, Bridge
1	Physical	Bit	Raw bit transmission	Hub, Repeater

5.1.4 Data Encapsulation in OSI

Key Concept: Encapsulation

As data travels *down* the OSI stack at the sender, each layer adds its own **header** (and sometimes a **trailer**). At the receiver, each layer strips the corresponding header as data travels *up* the stack. This process is called **encapsulation** / **de-encapsulation**.



5.2 The TCP/IP Reference Model

Definition: TCP/IP Model

The **TCP/IP model** (also called the *Internet model* or *DoD model*) is a 4-layer conceptual framework that was developed for the ARPANET and became the foundation of the modern Internet. It is a *practical* model based on protocols actually deployed.

5.2.1 Layers of the TCP/IP Model

Layer 4 – Application Layer Combines the OSI Application, Presentation, and Session layers. Contains all higher-level protocols for user applications.

Protocols: HTTP, FTP, SMTP, DNS, Telnet, SSH, SNMP, DHCP, POP3.

Layer 3 – Transport Layer Corresponds directly to the OSI Transport layer.

- **TCP:** Connection-oriented, reliable, ordered, congestion and flow controlled. Used by HTTP, FTP, SMTP.
- **UDP:** Connectionless, unreliable, fast, low overhead. Used by DNS, VoIP, video streaming.

Layer 2 – Internet Layer Corresponds to the OSI Network layer. Core protocol is IP.

Protocols: IPv4, IPv6, ICMP, ARP, RARP, IGMP.

Function: Logical addressing, routing, fragmentation.

Layer 1 – Network Access Layer Also called the *Link Layer* or *Host-to-Network layer*. Combines OSI Data Link and Physical layers.

Protocols: Ethernet, Wi-Fi, PPP, Frame Relay, ATM.

Function: Physical transmission, MAC addressing, framing.

5.2.2 TCP/IP Layer Summary

Table 5.2: TCP/IP Model – Layer Summary

Layer	Name	Key Protocols	PDU
4	Application	HTTP, FTP, SMTP, DNS, SSH, SNMP	Message
3	Transport	TCP, UDP	Segment / Datagram
2	Internet	IP (v4/v6), ICMP, ARP, IGMP	Packet
1	Network Access	Ethernet, Wi-Fi, PPP	Frame / Bit

6 Comparison of OSI and TCP/IP Reference Models

Exam Tip

The comparison of OSI and TCP/IP is a **high-yield exam topic**. Memorize the structural, functional, and philosophical differences.

6.1 Layer Mapping

Table 6.1: OSI vs. TCP/IP Layer Correspondence

OSI Model (7 layers)	TCP/IP Model (4 layers)
7 Application	4 Application
6 Presentation	
5 Session	
4 Transport	3 Transport
3 Network	2 Internet
2 Data Link	1 Network Access
1 Physical	

6.2 Detailed Comparison

Table 6.2: Detailed OSI vs. TCP/IP Comparison

Aspect	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO (International Organization for Standardization), 1984	US DOD / ARPA, 1970s–1980s
Number of layers	7	4
Nature	Conceptual / generic reference model	Practical / implementation model
Approach	Protocol-independent; defines services first, then protocols	Protocol-specific; model was defined after protocols were designed
Service interfaces	Clearly defines service, interface, and protocol	Boundaries not clearly defined
Transport layer	Guarantees delivery; both connection-oriented and connectionless	Both TCP (reliable) and UDP (unreliable); does not guarantee delivery at model level
Session/Presentation	Dedicated layers (5 & 6)	Merged into Application layer
Network layer	Both connection-oriented and connectionless at Network layer	Connectionless only at Internet layer
Data Link / Physical	Two separate layers	Merged into single Network Access layer
Protocols	Model defines protocols (not tied to specific ones)	Built around TCP, IP, UDP, etc.
Adoption	Widely taught but not deployed as-is	Foundation of the Internet; universally deployed
Flexibility	More structured; easier to replace one layer	Replacing layers is harder due to tight coupling of protocols
Overhead	7-layer processing can add overhead	Fewer layers; leaner processing

6.3 Similarities

- Both are **layered architectures** – each layer has defined responsibilities.
- Both use **encapsulation**: each layer adds headers to data as it passes down.
- Both have a **Transport layer** that provides end-to-end communication.
- Both have a **Network/Internet layer** for logical addressing and routing.
- Both are **open standards** (publicly available, not proprietary).
- Both support **connection-oriented and connectionless services** (at the transport layer at minimum).

6.4 Criticisms

OSI Model – Criticisms

- Some layers (Session, Presentation) are rarely used in practice.
- Overly complex for many applications.
- Arrived too late; TCP/IP was already dominant.
- Implementation was more complex than anticipated.

TCP/IP Model – Criticisms

- Does not clearly separate the concepts of service, interface, and protocol.
- Does not distinguish physical and data link layers.
- Not general enough to describe non-TCP/IP networks.
- The Network Access layer is vague and underspecified.

7 Important Protocols – Quick Reference

Table 7.1: Key Protocols at Each TCP/IP Layer

Layer	Protocol	Purpose
Application	HTTP / HTTPS	Web browsing; HTTPS adds TLS encryption
	FTP	File transfer (control: port 21, data: port 20)
	SMTP / POP3 / IMAP	Email send / receive
	DNS	Resolves domain names to IP addresses
	DHCP	Dynamically assigns IP addresses to hosts
	SSH	Encrypted remote terminal access (port 22)
	Telnet	Unencrypted remote terminal (port 23)
Transport	SNMP	Network device management and monitoring
	TCP	Reliable, ordered, connection-oriented delivery
Internet	UDP	Unreliable, fast, connectionless delivery
	IPv4	32-bit addressing; primary Internet protocol
	IPv6	128-bit addressing; next-generation Internet
	ICMP	Error reporting and diagnostics (ping, traceroute)
	ARP	Resolves IP addresses to MAC addresses
Network Access	RARP	Resolves MAC addresses to IP (legacy)
	Ethernet (802.3)	Dominant wired LAN technology
	Wi-Fi (802.11)	Wireless LAN
	PPP	Point-to-point WAN links

8 Practice Questions (MAKAUT Style)

Short Answer Questions (2–3 marks)

- Q1.** Define a computer network. State its goals.
- Q2.** What is the difference between LAN, MAN, and WAN?
- Q3.** What is a network topology? List the types.
- Q4.** State two advantages and two disadvantages of star topology.
- Q5.** What is the full form of OSI? How many layers does it have?
- Q6.** Name the PDU at each layer of the OSI model.
- Q7.** What is encapsulation in networking?
- Q8.** What protocol is used to resolve IP addresses to MAC addresses?

Long Answer Questions (5–6 marks)

- Q1.** Explain the seven layers of the OSI reference model with the function of each layer and the PDU at each layer.
- Q2.** Compare and contrast the OSI model and the TCP/IP model with a neat diagram.
- Q3.** Describe the Star, Bus, Ring, and Mesh network topologies. Draw neat diagrams for each and compare them in terms of cost, fault tolerance, and performance.
- Q4.** A full mesh network has 8 nodes. Calculate the number of links required and the number of ports needed per node.
- Q5.** Describe the functions of the following network devices: Hub, Switch, Router, and Gateway. At which OSI layer does each operate?

Numerical Answer

For the 8-node full mesh question above:

$$\text{Links} = \frac{8 \times 7}{2} = \mathbf{28 \text{ links}}$$

$$\text{Ports per node} = 8 - 1 = \mathbf{7 \text{ ports}}$$

9 Module 1 – Key Takeaways

Summary Checklist

- ✓ A network interconnects nodes for data sharing and communication.
- ✓ Key hardware: NIC, hub, switch, router, bridge, gateway.
- ✓ LAN (building) $\subset$ MAN (city) $\subset$ WAN (global).
- ✓ Bus: shared medium; Star: central device; Ring: token passing; Mesh: every-to-every.
- ✓ Mesh links formula: $n(n - 1)/2$.
- ✓ OSI has 7 layers; TCP/IP has 4 layers.
- ✓ OSI is a reference model; TCP/IP is implementation-based.
- ✓ Data encapsulation: Application $\rightarrow$ Physical at sender; reverse at receiver.
- ✓ Key protocols: HTTP (80), FTP (21/20), SMTP (25), DNS (53), SSH (22), Telnet (23).
- ✓ TCP = reliable, ordered; UDP = fast, unreliable.

Module 2 — The Physical Layer

Transmission Media, Encoding Schemes, Multiplexing & Switching

10 Hours · 12 Marks

1 Data Transmission Concepts

1.1 Data, Signal, and Transmission

Definition: Data

Data refers to raw information in the form of text, numbers, images, audio, or video that needs to be communicated from one entity to another.

Definition: Signal

A **signal** is the physical representation of data, encoded as variations in an electromagnetic or optical wave (voltage, current, light intensity, etc.) suitable for transmission over a medium.

Definition: Transmission

Transmission is the process of propagating and processing signals from one node to another through a communication medium.

1.2 Analog vs. Digital Data and Signals

Table 1.1: Analog vs. Digital – Data and Signals

Property	Analog	Digital
Nature	Continuous waveform; infinite values	Discrete levels; finite values (0 and 1)
Representation	Sine wave (amplitude, frequency, phase)	Square wave (voltage levels: high/low)
Noise immunity	Low – noise corrupts continuously	High – easy to regenerate 0/1
Bandwidth	Occupies a range of frequencies	Defined in bps (bit rate)
Attenuation	Signal weakens with distance	Regenerated by repeaters
Examples	Human voice, AM/FM radio	Computer data, digital telephone

1.3 Key Data Transmission Concepts

1.3.1 Bandwidth

Definition: Bandwidth

Bandwidth is the range of frequencies that a transmission medium can carry (analog: measured in Hz) or the maximum data rate a channel can support (digital: measured in bits per second, bps).

1.3.2 Data Rate and Bit Rate

Definition: Bit Rate

The **bit rate** is the number of bits transmitted per second (bps). It is determined by the bandwidth of the channel and the encoding scheme used.

1.3.3 Baud Rate vs. Bit Rate

Definition: Baud Rate

The **baud rate** (symbol rate) is the number of *signal changes* (symbols) per second. One baud may carry multiple bits if multiple signal levels are used.

Formula: Bit Rate vs. Baud Rate

$$\text{Bit Rate} = \text{Baud Rate} \times \log_2 M$$

where M is the number of distinct signal levels (modulation levels).

Example: If baud rate = 1000 symbols/s and $M = 4$ levels:

$$\text{Bit Rate} = 1000 \times \log_2 4 = 1000 \times 2 = 2000 \text{ bps}$$

1.3.4 Nyquist's Theorem (Noiseless Channel)

Formula: Nyquist Theorem

For a noiseless channel of bandwidth B Hz and M discrete signal levels:

$$\text{Maximum Data Rate} = 2B \log_2 M \quad \text{bps}$$

Example: $B = 3000$ Hz, $M = 8$:

$$C = 2 \times 3000 \times \log_2 8 = 6000 \times 3 = 18,000 \text{ bps}$$

1.3.5 Shannon's Capacity Theorem (Noisy Channel)

Formula: Shannon–Hartley Theorem

The theoretical maximum data rate of a noisy channel:

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \quad \text{bps}$$

where B = bandwidth (Hz), S/N = signal-to-noise ratio (linear scale).

Example: $B = 3000$ Hz, $S/N = 30$ dB $\Rightarrow S/N_{\text{linear}} = 1000$:

$$C = 3000 \times \log_2(1 + 1000) \approx 3000 \times 9.97 \approx 29,900 \text{ bps}$$

Note

SNR (dB) = $10 \log_{10}(S/N)$, so 30 dB $\Rightarrow S/N = 10^3 = 1000$. Shannon's theorem gives an *upper bound*; practical systems achieve less.

1.3.6 Transmission Modes

Table 1.2: Transmission Modes

Mode	Description	Example
Simplex	One direction only; sender cannot receive	TV broadcast, keyboard to CPU
Half-Duplex	Both directions, but not simultaneously	Walkie-talkie, older Ethernet (hub)
Full-Duplex	Both directions simultaneously	Telephone, modern switched Ethernet

1.3.7 Attenuation, Distortion, and Noise

- **Attenuation:** Loss of signal energy (amplitude reduction) as it travels through the medium. Compensated by *amplifiers* (analog) or *repeaters* (digital).
- **Distortion:** Change in the shape of the signal due to different propagation speeds at different frequencies.
- **Noise:** Random, unwanted signals superimposed on the transmitted signal. Types: thermal, induced, crosstalk, impulse.

2 Transmission Media

Definition: Transmission Medium

A **transmission medium** is the physical path between transmitter and receiver through which signals are propagated. Media are classified as **guided** (bounded, wired) or **unguided** (unbounded, wireless).

2.1 Guided (Wired) Transmission Media

2.1.1 Twisted Pair Cable

Key Concept: Twisted Pair Cable

Twisted pair consists of two insulated copper wires twisted together. The twisting reduces *electromagnetic interference (EMI)* and *crosstalk* by causing the noise induced on each wire to cancel out.

Types of Twisted Pair:

Table 2.1: UTP vs. STP Comparison

Property	UTP (Unshielded Twisted Pair)	STP (Shielded Twisted Pair)
Shielding	No metallic shield	Metallic foil/braid shield around pairs
EMI resistance	Moderate	High
Cost	Low (most common)	Higher
Installation	Easy, flexible	More complex
Bandwidth	Up to 10 Gbps (Cat 6a/7)	Up to 10 Gbps+
Use	Home/office LAN (Ethernet)	Industrial, high-interference environments

UTP Categories (EIA/TIA 568):

Table 2.2: UTP Cable Categories

Category	Max Frequency	Application
Cat 3	16 MHz	10BASE-T Ethernet, telephone
Cat 5	100 MHz	100BASE-TX (Fast Ethernet)
Cat 5e	100 MHz	1000BASE-T (Gigabit Ethernet)
Cat 6	250 MHz	10GBASE-T (up to 55 m)
Cat 6a	500 MHz	10GBASE-T (up to 100 m)
Cat 7	600 MHz	10 Gbps with full shielding

Advantages of Twisted Pair:

- Inexpensive and lightweight; easy to install
- Supports both analog and digital signals
- Widely available; standard for telephone and LAN

Disadvantages of Twisted Pair:

- Limited bandwidth and distance compared to other media
- Susceptible to EMI (especially UTP)
- Attenuation increases with distance and frequency

2.1.2 Coaxial Cable**Key Concept: Coaxial Cable**

Coaxial cable has a central solid copper conductor surrounded by a plastic insulator, a metallic braid/foil shield, and an outer plastic jacket. The shielding provides excellent EMI protection and wider bandwidth than twisted pair.

Structure (from center outward):

1. Inner conductor (solid or stranded copper)
2. Dielectric insulator (polyethylene)
3. Metallic shield (braided copper or aluminum foil)
4. Outer plastic jacket (PVC)

Types:

- **Baseband coax** (50 Ω): Single digital signal at a time. Used in 10BASE2 (Thinnet) and 10BASE5 (Thicknet) Ethernet.
- **Broadband coax** (75 Ω): Multiple analog signals via FDM. Used in cable TV (CATV) distribution.

Table 2.3: Coaxial Cable – Advantages and Disadvantages

Advantages	Disadvantages
Higher bandwidth than twisted pair	More expensive than twisted pair
Better EMI shielding	Bulkier and less flexible
Longer distances possible	Single point of failure (bus coax)
Supports both analog and digital	Largely replaced by fiber/UTP

2.1.3 Optical Fiber Cable

Key Concept: Optical Fiber

Optical fiber transmits data as pulses of *light* through a glass or plastic core using the principle of **total internal reflection**. It offers the highest bandwidth and longest distances of all transmission media.

Structure:

1. **Core:** Ultra-pure glass or plastic; carries the light signal.
2. **Cladding:** Glass/plastic layer with a lower refractive index than the core; causes total internal reflection.
3. **Buffer coating:** Plastic jacket protecting the fiber.
4. **Strength members and outer jacket:** Physical protection.

Total Internal Reflection: When light travels from a denser medium (core, refractive index n_1) to a less dense medium (cladding, $n_2 < n_1$) at an angle greater than the *critical angle* θ_c :

$$\theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$$

Light is completely reflected back into the core and propagates along the fiber.

Types of Optical Fiber:

Table 2.4: Single-Mode vs. Multi-Mode Fiber

Property	Single-Mode Fiber (SMF)	Multi-Mode Fiber (MMF)
Core diameter	~8–10 μm	50 or 62.5 μm
Light source	Laser diode	LED
Bandwidth	Very high (100+ Gbps)	Moderate (up to 10 Gbps)
Distance	Up to 100+ km	Up to 2 km (OM4)
Dispersion	Very low	Higher (modal dispersion)
Cost	Higher	Lower
Use case	Long-haul WAN, backbone	Data centers, campus LAN

Table 2.5: Optical Fiber – Advantages and Disadvantages

Advantages	Disadvantages
Extremely high bandwidth (Tbps)	Expensive to install and splice
Very low attenuation (long distances)	Fragile; difficult to work with
Immune to EMI and crosstalk	Unidirectional (needs two fibers for full-duplex)
Secure (hard to tap without detection)	Specialized connectors and test equipment
Lightweight and thin	

Exam Tip

Remember the key comparison: Twisted Pair (cheapest, shortest range) < Coaxial (medium) < Optical Fiber (fastest, longest range, most expensive).

2.2 Unguided (Wireless) Transmission Media

Definition: Unguided Media

Unguided media (also called *wireless* or *unbounded* media) transmit electromagnetic signals through free space (air, vacuum) without a physical conductor. The signals propagate through the atmosphere or outer space.

2.2.1 Radio Waves

Key Concept: Radio Waves

Radio waves span frequencies from approximately **3 kHz to 1 GHz**. They are *omnidirectional* (propagate in all directions) and can penetrate walls and obstacles.

- **Propagation modes:**
 - *Ground wave* (below 2 MHz): Follows Earth's curvature.
 - *Sky wave* (2–30 MHz): Reflects off the ionosphere for long-distance communication.
 - *Line-of-sight* (above 30 MHz): Direct path between transmitter and receiver.
- **Applications:** AM/FM radio, cordless phones, cellular networks (2G/3G), Bluetooth (2.4 GHz ISM band).
- **Standards:** IEEE 802.11 (Wi-Fi) uses the 2.4 GHz and 5 GHz radio bands.

Advantages: Long range, penetrates walls, omnidirectional (no antenna alignment needed).

Disadvantages: Susceptible to interference, limited bandwidth at lower frequencies, regulated by government (ITU, FCC).

2.2.2 VHF (Very High Frequency)

VHF spans **30 MHz to 300 MHz**.

- Propagates primarily by line-of-sight.
- Used for: FM radio (88–108 MHz), VHF TV broadcasts, aviation communication (108–137 MHz), marine radio.
- Better bandwidth than lower-frequency radio; limited by line-of-sight range (typically 100–200 km).

2.2.3 Microwave Transmission

Key Concept: Microwave

Microwaves occupy the frequency range **1 GHz to 300 GHz**. They are *unidirectional* (highly directional, narrow beams) and require *line-of-sight* alignment between transmitter and receiver antennas.

Types:

1. Terrestrial Microwave:

- Uses dish antennas on towers, mountains, or tall buildings.
- Repeater stations needed every 50–80 km (due to Earth's curvature).
- Used for telephone trunk lines, TV distribution, cellular backhaul.

2. Satellite Microwave:

- Uses a satellite in orbit as a relay station.
- Uplink (ground to satellite) and downlink (satellite to ground).
- See satellite section below.

Advantages: No need for physical cables, high bandwidth, covers long distances.

Disadvantages: Line-of-sight required, affected by rain fade and atmospheric absorption, expensive towers/equipment.

2.2.4 Satellite Communication

Key Concept: Satellite

A communications **satellite** is a microwave relay station placed in orbit. It receives signals from ground stations (uplink), amplifies them, and retransmits them to other ground stations (downlink) on a different frequency.

Types of Satellite Orbits:

Table 2.6: Satellite Orbit Types

Type	Altitude	Latency	Application
GEO (Geostationary)	35,786 km	~270 ms (one-way)	TV broadcast, weather, traditional broadband
MEO (Medium Earth Orbit)	2,000–20,000 km	50–150 ms	GPS, navigation systems
LEO (Low Earth Orbit)	160–2,000 km	20–40 ms	Starlink, Iridium, Teledesic

Note

GEO satellites appear *stationary* over the equator because their orbital period (24 hours) matches Earth's rotation. One GEO satellite can cover approximately **1/3 of Earth's surface**. Three GEO satellites can achieve near-global coverage (excluding polar regions).

Advantages: Very large coverage area, independent of terrestrial infrastructure, good for broadcasting.

Disadvantages: High propagation delay (especially GEO), high launch cost, weather/rain fade, signal attenuation.

2.2.5 Infrared (IR) Transmission

Key Concept: Infrared

Infrared occupies frequencies from approximately **300 GHz to 400 THz** (wavelengths 1 mm to 750 nm). It is used for short-range, indoor wireless communication.

- Does *not* penetrate walls – naturally contained within a room (enhances security).
- No government licensing required.
- Used in: TV remote controls, IrDA (Infrared Data Association) for device-to-device links, short-range point-to-point links.
- **Disadvantages:** Very short range (typically < 1 m to a few meters), blocked by obstacles, affected by sunlight interference.

2.3 Guided vs. Unguided Media – Summary

Table 2.7: Overall Comparison of Transmission Media

Medium	Bandwidth	Range	Best Use
Twisted Pair (UTP)	Up to 10 Gbps	Up to 100 m	LAN, telephone
Coaxial Cable	500 MHz	km-range	Cable TV, legacy LAN
Optical Fiber	100+ Gbps	100+ km	WAN backbone, high-speed LAN
Radio Waves	kHz – 1 GHz	km to global	Broadcasting, cellular
VHF	30–300 MHz	100–200 km	FM radio, TV, aviation
Microwave	1–300 GHz	50–80 km (terrestrial)	Telephone trunk, backhaul
Satellite	GHz range	Global	TV broadcast, GPS, broadband
Infrared	THz range	<1 m – few m	Remote control, IrDA

3 Encoding Schemes

Definition: Encoding

Encoding (line coding) is the process of converting digital data (bits) into digital signals for transmission. The choice of encoding scheme affects bandwidth efficiency, synchronization, error detection, and DC bias.

Key properties of a good encoding scheme:

- **Self-clocking:** Signal transitions allow the receiver to recover the clock (synchronization).
- **No DC component:** DC-balanced signals pass through transformers and AC-coupled links.
- **Error detection capability:** Some schemes allow detection of certain errors.
- **Bandwidth efficiency:** Minimum bandwidth for a given bit rate.

3.1 NRZ (Non-Return-to-Zero)

Key Concept: NRZ Encoding

In **NRZ encoding**, the signal does not return to zero between bits. The voltage level directly represents the bit value throughout the bit interval.

NRZ-L (NRZ-Level):

- Bit 1 → High voltage (+V)
- Bit 0 → Low voltage (0 or −V)
- **Problem:** Long strings of 0s or 1s cause loss of synchronization (no transitions) and a DC component.

NRZ-I (NRZ-Invert on ones):

- Bit 1 → *Change* (invert) the signal level
- Bit 0 → *No change* in signal level
- **Improvement:** Solves the problem of long strings of 1s; still has issues with long strings of 0s.

Table 3.1: NRZ Encoding – Advantages and Disadvantages

Advantages	Disadvantages
Simple to implement	Synchronization problems with long runs of same bit
Bandwidth efficient (1 bit per baud)	DC component present (NRZ-L)
Easy hardware	No error detection

3.2 Manchester Encoding

Key Concept: Manchester Encoding

Manchester encoding (IEEE 802.3) uses a *mid-bit transition* to represent each bit. This provides **self-clocking** (the receiver can synchronize from the signal itself).

Convention (IEEE 802.3):

- Bit 1 → **High-to-Low** transition at mid-bit (falling edge in mid-period)
- Bit 0 → **Low-to-High** transition at mid-bit (rising edge in mid-period)

Note

Alternative convention (G.E. Thomas): 1 = Low-to-High, 0 = High-to-Low. Both are used in different standards – check which is specified.

Characteristics:

- There is always a transition at the *middle* of each bit period.

- There may or may not be a transition at the boundary between bits.
- **No DC component** – balanced encoding.
- **Self-synchronizing** – clock can always be recovered.
- **Drawback:** Requires *twice the bandwidth* of NRZ (the baud rate is double the bit rate).

Table 3.2: Manchester Encoding – Advantages and Disadvantages

Advantages	Disadvantages
Self-clocking (no sync loss)	Requires double the bandwidth of NRZ
No DC component	Lower bandwidth efficiency
Simple error detection possible	

Used in: 10 Mbps Ethernet (10BASE-T, 10BASE2), 10BASE5.

3.3 Differential Manchester Encoding

Key Concept: Differential Manchester

Differential Manchester encoding (IEEE 802.5, Token Ring) uses the *presence or absence of a transition at the beginning* of each bit period to encode data, while always maintaining a mid-bit transition for clocking.

Convention:

- Bit **0** → Transition *at the beginning* of the bit period (in addition to the mandatory mid-bit transition).
- Bit **1** → *No transition* at the beginning of the bit period (only the mandatory mid-bit transition occurs).

Characteristics:

- **Always** has a mid-bit transition for synchronization.
- Encoding is **differential** – each bit's value depends on the transition relative to the previous state, not the absolute voltage level.
- More **noise immune** than Manchester (polarity errors do not affect decoding).
- Used in: **Token Ring** (IEEE 802.5), ISDN.

3.4 Encoding Scheme Comparison

Table 3.3: Comparison of NRZ, Manchester, and Differential Manchester

Property	NRZ	Manchester	Differential Manchester
Transition at mid-bit	No	Always	Always
Transition at bit boundary	Possible	Possible	Depends on bit value
DC component	Yes (NRZ-L)	No	No
Self-clocking	No	Yes	Yes
Bandwidth efficiency	High (1 bit/ baud)	Low (2 baud/ bit)	Low (2 baud/ bit)
Noise immunity	Low	Moderate	High
Standard	USB, HDMI	10BASE-T Ethernet	Token Ring (802.5)

Exam Tip

Draw the waveform for the bit pattern 10110010 for each encoding scheme. This is a classic MAKAUT question. Memorize:

- NRZ-L: level directly represents bit.
- Manchester: mid-bit transition always; 1 = falling, 0 = rising (IEEE).
- Diff. Manchester: mid-bit always; 0 = boundary transition, 1 = no boundary transition.

4 Multiplexing Techniques

Definition: Multiplexing

Multiplexing is the technique of combining multiple signals from different sources and transmitting them simultaneously over a single high-capacity communication channel (link). A **multiplexer (MUX)** combines signals at the sender; a **demultiplexer (DEMUX)** separates them at the receiver.

Why multiplex?

- Transmission media (especially fiber, satellite) have far greater bandwidth than a single signal requires.
- Sharing a link reduces cost.
- Increases channel utilization efficiency.

4.1 Frequency Division Multiplexing (FDM)

Key Concept: FDM

Frequency Division Multiplexing (FDM) divides the total available bandwidth of the channel into *non-overlapping frequency sub-bands* (channels). Each signal is modulated onto a different *carrier frequency*, and all signals are transmitted simultaneously.

How it works:

1. Each input signal is modulated (AM/FM) onto a unique carrier frequency $f_1, f_2, \dots, f_n$.
2. The modulated signals are combined and sent over the shared link.
3. At the receiver, bandpass filters separate each signal by its frequency band, then demodulate.

Guard Bands: Small unused frequency gaps between sub-channels to prevent inter-channel interference (crosstalk).

Table 4.1: FDM – Advantages and Disadvantages

Advantages	Disadvantages
All channels transmitted simultaneously	Wasted bandwidth in guard bands
Simple; no timing coordination needed	Intermodulation distortion possible
Proven technology (decades of use)	Not suitable for bursty/digital sources

Applications: AM/FM radio broadcasting, cable TV (CATV), traditional telephone (PSTN), ADSL.

Example 4.1. A 1 MHz channel can support 10 phone calls each using 100 kHz, with 10 kHz guard bands between each pair:

$$10 \times 100 \text{ kHz} + 9 \times 10 \text{ kHz} = 1000 + 90 = 1090 \text{ kHz} \approx 1 \text{ MHz}$$

4.2 Time Division Multiplexing (TDM)

Key Concept: TDM

Time Division Multiplexing (TDM) divides the transmission time of the channel into *fixed time slots*. Each signal is assigned a dedicated slot in a repeating *frame*. The full bandwidth is available to each signal, but only during its assigned time slot.

TDM Frame Structure:

$$\underbrace{[S_1|S_2|S_3|\dots|S_n]}_{\text{Frame}} \underbrace{[S_1|S_2|S_3|\dots|S_n]}_{\text{Frame}} \dots$$

Each slot S_i carries one sample (or group of bits) from source i .

Types of TDM:

1. **Synchronous TDM:** Slots are pre-assigned to sources regardless of whether the source has data to send. Simple but wastes bandwidth if sources are idle.
2. **Statistical (Asynchronous) TDM:** Slots are assigned *on demand* to sources that have data. More efficient but requires overhead for addressing.

Table 4.2: TDM – Advantages and Disadvantages

Advantages	Disadvantages
Full channel bandwidth available each turn	Synchronous TDM wastes slots when idle
Suitable for digital data	Requires tight synchronization
Simple demultiplexing (time-based)	Higher speed required at the link

Applications: T1/E1 telephone systems, SONET/SDH, ISDN, DS hierarchy.

Formula: TDM Data Rate

If n sources each have a bit rate of R bps, the composite link must carry:

$$R_{\text{link}} = n \times R \text{ bps}$$

Example 4.2. Four 1 Mbps channels are multiplexed using TDM.

Link data rate = $4 \times 1 \text{ Mbps} = 4 \text{ Mbps}$.

Each source gets $1/4$ of the frame time, receiving its full 1 Mbps within that slot.

T-Carrier System (North America):

Table 4.3: T-Carrier Hierarchy

Level	Channels	Bit Rate
T1 (DS1)	24 DS0 channels	1.544 Mbps
T2 (DS2)	4 T1s (96 DS0)	6.312 Mbps
T3 (DS3)	28 T1s (672 DS0)	44.736 Mbps
T4 (DS4)	168 T1s (4032 DS0)	274.176 Mbps

4.3 Wavelength Division Multiplexing (WDM)

Key Concept: WDM

Wavelength Division Multiplexing (WDM) is the *optical analog of FDM*. Multiple optical signals at different **wavelengths** (colors of light) are combined and transmitted simultaneously over a single optical fiber.

How it works:

1. Multiple laser sources transmit at different wavelengths $(\lambda_1, \lambda_2, \dots, \lambda_n)$.

2. An optical multiplexer (prism or diffraction grating) combines them onto one fiber.
3. At the receiver, an optical demultiplexer separates the wavelengths.
4. Each wavelength channel can carry independent data streams.

Types:

- **CWDM** (Coarse WDM): 18 wavelengths, wider channel spacing (20 nm), less expensive.
- **DWDM** (Dense WDM): Up to 160 wavelengths, narrow channel spacing (≤ 0.8 nm), uses erbium-doped fiber amplifiers (EDFAs). Can achieve **multiple terabits per second** on a single fiber.

Table 4.4: Comparison of FDM, TDM, and WDM

Property	FDM	TDM	WDM
Medium	Analog/RF	Digital (any)	Optical fiber only
Separation basis	Frequency	Time slots	Wavelength (color)
Signal type	Analog	Digital	Optical
Guard bands needed	Yes	No (guard time)	No (wavelength isolation)
Capacity	Moderate	High	Extremely high (Tbps)
Synchronization needed	No	Yes	No
Application	Radio, cable TV	T1/E1, SONET	Fiber backbone, DWDM

5 Switching Techniques

Definition: Switching

Switching is the mechanism by which a connection (or data) is routed from a source to a destination through a network of intermediate nodes. Different switching techniques offer different trade-offs in delay, efficiency, and reliability.

The three fundamental switching techniques are:

1. Circuit Switching
2. Message Switching
3. Packet Switching

5.1 Circuit Switching

Key Concept: Circuit Switching

In **circuit switching**, a *dedicated, physical end-to-end path* is established between source and destination *before* data transmission begins. This path is reserved for the exclusive use of the communication session until it is explicitly released.

Three phases of circuit switching:

1. **Circuit Establishment:** A signaling message is propagated from source to destination, reserving resources (bandwidth, switches) at each intermediate node. A dedicated path is established.
2. **Data Transfer:** Data is transmitted continuously over the established circuit. The full reserved bandwidth is available, and there is no addressing overhead per data unit.
3. **Circuit Teardown (Disconnection):** When communication ends, a disconnect signal releases all reserved resources along the path.

Formula: Circuit Switching Total Time

$$T_{\text{total}} = T_{\text{setup}} + T_{\text{transmission}} + T_{\text{teardown}}$$

where $T_{\text{transmission}} = \frac{\text{Data size (bits)}}{\text{Bandwidth (bps)}}$.

Table 5.1: Circuit Switching – Advantages and Disadvantages

Advantages	Disadvantages
Guaranteed bandwidth and QoS	Resources wasted during idle periods
Constant, predictable delay (real-time)	Setup time before data can flow
Simple data transfer (no per-packet routing)	Not suitable for bursty data traffic
Suitable for voice (PSTN)	Blocking: call rejected if resources unavailable

Example: Traditional **PSTN** (Public Switched Telephone Network), ISDN, dedicated leased lines.

5.2 Message Switching

Key Concept: Message Switching

In **message switching**, the entire message is treated as a single unit. Each intermediate node (switch) receives the *complete message*, stores it in its memory, and then *forwards* it to the next node when the outgoing link is available. This is the **store-and-forward** model.

How it works:

1. Source sends the complete message to the first switch.
2. Each switch stores the entire message, then forwards it to the next switch.
3. The message arrives at the destination in its entirety after traversing all hops.

Table 5.2: Message Switching – Advantages and Disadvantages

Advantages	Disadvantages
No dedicated circuit needed	Very long delays for large messages
Links shared efficiently	Intermediate nodes need large storage
Messages queued during congestion	High queuing and processing delay
Priority can be assigned to messages	Entire message lost if a node fails

Example: Telex, early email systems, telegraph networks.

Note

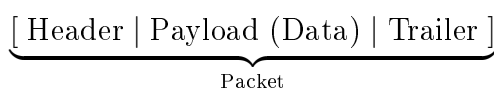
Message switching is largely **obsolete** in modern networks. It was a stepping stone to packet switching. The Internet uses packet switching.

5.3 Packet Switching

Key Concept: Packet Switching

In **packet switching**, the message is divided into smaller units called **packets**. Each packet is transmitted *independently* through the network and may take *different paths* to the destination. Packets are reassembled at the destination.

Packet structure:



The header contains source/destination addresses, sequence number, and other control information.

Types of Packet Switching:

1. **Datagram Switching (Connectionless):**

- Each packet is routed *independently*; packets of the same message may take different paths.
- Packets may arrive *out of order*; receiver reorders them.
- Each packet carries full addressing information.
- Used in: **IP (Internet Protocol)**, the Internet.
- Robust: if a node fails, packets are re-routed automatically.

2. Virtual Circuit Switching (Connection-Oriented):

- A *logical path* (virtual circuit) is established before transmission. All packets follow this same path.
- Packets arrive *in order*.
- Combines benefits of circuit switching (ordered delivery) and packet switching (resource sharing).
- Used in: **ATM**, Frame Relay, MPLS.

Formula: Packet Switching Delay

For store-and-forward packet switching with k hops, n packets of size L bits each, and link bandwidth R bps, ignoring propagation delay:

$$T = (k + n - 1) \times \frac{L}{R}$$

Compare with circuit switching (which requires setup but then transmits the entire message at line rate):

$$T_{\text{circuit}} \approx T_{\text{setup}} + \frac{nL}{R}$$

Packet switching achieves **pipelining**: switches forward packets in parallel, dramatically reducing overall delay.

Table 5.3: Packet Switching – Advantages and Disadvantages

Advantages	Disadvantages
Efficient use of bandwidth (statistical multiplexing)	Variable delay (jitter) – bad for real-time
No call setup delay	Packet overhead (headers on each packet)
Robust: multiple paths possible	Out-of-order delivery (datagram)
Suitable for bursty data traffic	Reassembly complexity at destination

Example: The **Internet** (IP), TCP/IP networks, Frame Relay, ATM.

5.4 Switching Techniques – Master Comparison

Table 5.4: Comprehensive Comparison of Switching Techniques

Property	Circuit Switching	Message Switching	Packet Switching
Dedicated path	Yes – physical	No	No (datagram) / Logical (VC)
Resource reservation	Yes (before transfer)	No	No
Store-and-forward	No	Yes (entire message)	Yes (per packet)
Delay type	Setup delay then constant	Large variable delay	Small variable delay (jitter)
Bandwidth efficiency	Poor (idle waste)	Good	Best (statistical mux)
Suitable for	Voice, real-time	Store-and-forward apps	Internet, data networks
Out-of-order delivery	No	No	Possible (datagram)
Packet loss possible	No	Low (node buffers)	Yes (buffer overflow)
Examples	PSTN, ISDN	Telex, early email	Internet, ATM

6 Module 2 – Numerical Problems

Worked Numericals

Problem 1: A digital channel has a bandwidth of 4000 Hz. Using 8 signal levels, what is the maximum data rate (Nyquist)?

$$C = 2 \times 4000 \times \log_2 8 = 8000 \times 3 = \mathbf{24,000} \text{ bps}$$

Problem 2: A channel has bandwidth 4000 Hz and SNR = 30 dB. Find Shannon's capacity.

$$\text{SNR (linear)} = 10^{30/10} = 1000$$

$$C = 4000 \times \log_2(1001) \approx 4000 \times 9.97 = \mathbf{39,868} \text{ bps}$$

Problem 3: 24 voice channels, each 4 kHz, are multiplexed using FDM with 400 Hz guard bands between each channel. What is the total bandwidth?

$$B = 24 \times 4000 + 23 \times 400 = 96,000 + 9,200 = \mathbf{105,200} \text{ Hz}$$

Problem 4: 4 channels, each at 1 Mbps, are TDM-multiplexed. A 100-bit frame uses 1 bit for synchronization. What is the link rate and efficiency?

$$\text{Link rate} = 4 \times 1 \text{ Mbps} + \frac{1}{99} \times 4 \text{ Mbps} \approx \mathbf{4.04} \text{ Mbps}$$

Problem 5 (Packet Switching): A message of 10,000 bits is sent over 3 hops. It is divided into 5 packets of 2000 bits each. Link speed = 1 Mbps.

$$T = (3 + 5 - 1) \times \frac{2000}{10^6} = 7 \times 0.002 = \mathbf{14} \text{ ms}$$

Without packetizing: $T = 3 \times 10,000/10^6 = 30 \text{ ms}$. Packetizing is **2.14**× faster due to pipelining.

7 Practice Questions (MAKAUT Style)

Short Answer Questions (2–3 marks)

- Q1.** Define bandwidth. Distinguish between analog bandwidth and digital bandwidth.
- Q2.** State Nyquist's theorem and Shannon's capacity theorem.
- Q3.** What is the difference between guided and unguided transmission media?
- Q4.** Distinguish between UTP and STP cables.
- Q5.** What is total internal reflection? How does it help in optical fiber?
- Q6.** What are the advantages of optical fiber over twisted pair?
- Q7.** State two differences between FDM and TDM.
- Q8.** Define multiplexing. Why is it needed?
- Q9.** What is a guard band in FDM?
- Q10.** What is the difference between synchronous and statistical TDM?
- Q11.** Define circuit switching and state its main disadvantage.
- Q12.** What is store-and-forward switching?
- Q13.** What is the difference between datagram packet switching and virtual circuit switching?

Long Answer Questions (5–6 marks)

- Q1.** Explain the three types of guided transmission media with neat diagrams, advantages, disadvantages, and applications.
- Q2.** Compare and contrast NRZ, Manchester, and Differential Manchester encoding. Draw waveforms for the bit sequence 1 0 1 1 0 0 1 0.
- Q3.** Explain FDM, TDM, and WDM with diagrams. Compare them in terms of medium, efficiency, and application.
- Q4.** Compare circuit switching, message switching, and packet switching. Discuss the types of delays in each.
- Q5.** A noiseless channel has a bandwidth of 3000 Hz and can support 8 signal levels. Calculate the Nyquist data rate. If the channel has an SNR of 30 dB, calculate the Shannon capacity.
- Q6.** Explain the satellite communication system. Describe GEO, MEO, and LEO satellites with their advantages and disadvantages.

8 Module 2 – Key Takeaways

Summary Checklist

- ✓ Nyquist: $C = 2B \log_2 M$ (noiseless). Shannon: $C = B \log_2(1 + S/N)$ (noisy).
- ✓ Guided media: Twisted Pair < Coaxial < Optical Fiber (speed and range increase).
- ✓ Unguided: Radio (omnidirectional, penetrates walls) → Microwave (directional, LOS) → Satellite (global) → IR (room-only).
- ✓ NRZ: no mid-bit transition; Manchester: always mid-bit (IEEE: 1=falling); Diff. Manchester: mid-bit always, 0=boundary transition.
- ✓ FDM: frequency bands; TDM: time slots; WDM: wavelengths on fiber.
- ✓ Circuit: dedicated path, constant delay, wastes resources.
- ✓ Message: whole message stored and forwarded (obsolete).
- ✓ Packet: divided into packets, statistical multiplexing, pipelining; datagram (IP) or virtual circuit (ATM, MPLS).
- ✓ Packet switching formula: $T = (k + n - 1) \times L/R$ for k hops, n packets.

Module 3 — The Data Link Layer

*Flow Control, Error Detection/Correction, ARQ, HDLC, MAC,
IEEE 802*

11 Hours · 15 Marks

1 Data Link Layer – Issues and Overview

Definition: Data Link Layer

The **Data Link Layer** (OSI Layer 2) is responsible for reliable *node-to-node* (hop-to-hop) transfer of data frames over a single communication link. It takes raw bits from the Physical Layer and organizes them into structured **frames**, providing error detection, flow control, and medium access control.

1.1 Functions of the Data Link Layer

1. **Framing:** Divides the stream of bits from the Network Layer into discrete, manageable units called *frames*. Adds delimiters (flags/headers/trailers) so the receiver can identify frame boundaries.
2. **Physical Addressing (MAC):** Adds the source and destination *MAC addresses* to the frame header for hop-to-hop delivery.
3. **Error Detection and Correction:** Detects (and sometimes corrects) bit errors introduced by the Physical Layer using redundancy bits (parity, CRC, checksum, Hamming code).
4. **Flow Control:** Prevents a fast sender from overwhelming a slow receiver by regulating the rate of data transmission.
5. **Error Control (ARQ):** Ensures reliable delivery by re-transmitting lost or corrupted frames using acknowledgements and timers.
6. **Access Control (MAC):** When multiple devices share the same medium, the MAC sub-layer determines which device can transmit at any given time (CSMA/CD, ALOHA, token passing, etc.).
7. **Link Management:** Establishes, maintains, and releases the data link connection.

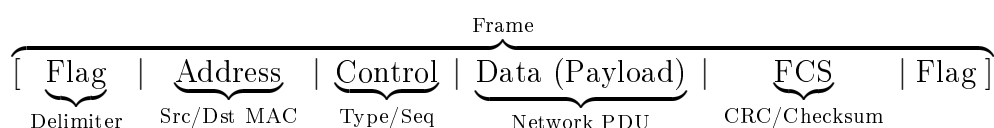
1.2 Data Link Layer Sub-layers

Table 1.1: IEEE 802 Data Link Sub-layers

Sub-layer	Function	Standard
LLC (Logical Link Control)	Error control, flow control, framing; interface to Network Layer	IEEE 802.2
MAC (Medium Access Control)	Physical addressing, frame delimiting, collision handling	IEEE 802.3/4/5/11

1.3 Frame Structure

A generic frame has the following fields:



Framing methods:

- **Character count:** A field in the header specifies the number of bytes in the frame. Simple but error-prone (a corrupted count loses sync).
- **Flag bytes with byte stuffing:** Special flag byte (e.g., 0x7E) marks start/end. If the data contains the flag byte, a special escape byte is inserted before it (*byte stuffing*).
- **Flag bits with bit stuffing:** A flag pattern (e.g., 01111110) marks boundaries. If five consecutive 1s appear in data, a 0 is *stuffed* after them. The receiver removes stuffed bits.
- **Physical layer coding violations:** Use illegal encoding symbols as delimiters.

2 Flow Control

Definition: Flow Control

Flow control is the mechanism by which the receiver controls the rate at which the sender transmits data, preventing the receiver's buffer from overflowing. It regulates the *amount of data* in transit between sender and receiver.

2.1 Stop-and-Wait Protocol

Key Concept: Stop-and-Wait

In **Stop-and-Wait**, the sender transmits **one frame** and then *stops* and waits for an **acknowledgement (ACK)** from the receiver before sending the next frame.

Operation:

1. Sender transmits Frame 0.
2. Sender starts a retransmission *timer*.
3. Receiver receives Frame 0 correctly; sends ACK 1 (ready for Frame 1).
4. Sender receives ACK 1; sends Frame 1. And so on.
5. If timer expires before ACK is received, sender *retransmits* the frame (timeout-based retransmission).

Sequence numbers: Only 1 bit needed (0 and 1 alternate). The sender uses the sequence number to detect duplicate frames (if an ACK is lost and the frame is retransmitted, the receiver can detect it is a duplicate).

Formula: Stop-and-Wait Efficiency

Let T_f = frame transmission time, T_p = propagation delay, T_a = ACK transmission time ≈ 0 .

$$a = \frac{T_p}{T_f} \quad (\text{propagation-to-transmission ratio})$$

$$\eta = \frac{1}{1 + 2a} \quad (\text{efficiency / utilization})$$

For high a (satellite links), efficiency is very poor.

Table 2.1: Stop-and-Wait – Advantages and Disadvantages

Advantages	Disadvantages
Simple to implement	Very low efficiency for long-delay links
Requires minimum buffer (1 frame)	Sender idle during propagation delay
Easy error control	Not suitable for high-speed, long-distance

2.2 Sliding Window Flow Control

Key Concept: Sliding Window Protocol

Sliding window allows the sender to transmit up to W frames (the *window size*) *without waiting* for an ACK. As ACKs are received, the window *slides forward*, allowing new frames to be sent. This keeps the channel busy (pipelined transmission).

Key parameters:

- **Sender window size W_s :** Maximum number of unacknowledged frames the sender may have outstanding at any time.
- **Receiver window size W_r :** Number of out-of-order frames the receiver can buffer.
- **Sequence number space:** With n bits, sequence numbers range from 0 to $2^n - 1$.

Formula: Sliding Window Efficiency

$$\eta = \begin{cases} \frac{W}{1+2a} & \text{if } W < 1+2a \\ 1 & \text{if } W \geq 1+2a \end{cases}$$

where W = window size, $a = T_p/T_f$.

Optimal window size to achieve 100% efficiency: $W_{\min} = 1 + 2a$ frames.

Example 2.1. A satellite link has $T_p = 270$ ms, $T_f = 10$ ms.

$$a = 270/10 = 27, \quad W_{\min} = 1 + 2(27) = 55 \text{ frames}$$

Stop-and-Wait efficiency: $\eta = 1/(1 + 54) = 1.8\%$.

With $W = 55$: $\eta = 55/55 = 100\%$.

Sliding window variants:

- **Go-Back-N:** $W_s \leq 2^n - 1$, $W_r = 1$.
- **Selective Repeat:** $W_s \leq 2^{n-1}$, $W_r = W_s$.

3 Error Detection Algorithms

Definition: Error Detection

Error detection is the process of detecting bit errors in a transmitted frame by adding *redundancy* (extra bits) to the data. The receiver recomputes the redundancy and compares it with the received redundancy to detect errors.

Types of errors:

- **Single-bit error:** Only one bit is changed ($0 \rightarrow 1$ or $1 \rightarrow 0$).
- **Burst error:** Two or more consecutive bits are changed. More common in real channels.

3.1 Parity Check

Key Concept: Parity Check

A **parity bit** is appended to a data word such that the total number of 1s in the codeword (data + parity) is either even (*even parity*) or odd (*odd parity*).

3.1.1 Simple (1D) Parity Check

- **Even parity:** Parity bit chosen so total 1s is even.
- **Odd parity:** Parity bit chosen so total 1s is odd.

- Detects all **single-bit errors** and any **odd number** of errors.
- **Cannot detect** even numbers of bit errors.
- **Cannot correct** any errors.

Worked Example: Simple Parity

Data: 1101011 (five 1s – odd count).

For **even parity**: append 1 $\Rightarrow$ codeword: 11010111 (six 1s, even).

For **odd parity**: append 0 $\Rightarrow$ codeword: 11010110 (five 1s, odd).

3.1.2 Two-Dimensional (2D) Parity Check

- Data is arranged in a **matrix** (rows and columns).
- A parity bit is computed for each *row* and each *column*.
- Can detect **all single-bit errors** and **most burst errors**.
- Can **correct single-bit errors** (the row and column parity pinpoints the exact bit).

3.2 Cyclic Redundancy Check (CRC)

Key Concept: CRC

CRC is the most powerful and widely used error-detection technique. It treats the data as a large binary polynomial and divides it by a pre-agreed *generator polynomial* $G(x)$. The remainder of this division (the *Frame Check Sequence, FCS*) is appended to the data. The receiver performs the same division; a non-zero remainder indicates an error.

CRC Algorithm (Sender):

1. Let message M have k bits and generator G have $r + 1$ bits (degree r).
2. Append r zeros to M to get M' (of length $k + r$ bits).
3. Divide M' by G using **modulo-2 division** (XOR, no carries/borrows).
4. The remainder R (r bits) is the **FCS**.
5. Transmit $T = M'$ with the last r bits replaced by R , i.e., $T = M \cdot 2^r \oplus R$ (i.e., M followed by R).

CRC Algorithm (Receiver):

1. Divide the received frame T' by G using modulo-2 division.
2. If remainder = 0: **no error detected**; accept frame.
3. If remainder $\neq 0$: **error detected**; discard/request retransmission.

Modulo-2 Division (XOR):

- $0 \oplus 0 = 0$, $1 \oplus 0 = 1$, $0 \oplus 1 = 1$, $1 \oplus 1 = 0$
- No carry or borrow – purely bitwise XOR.

Worked Example: CRC Calculation

Message $M = 1101011011$, Generator $G = 10011$ ($r = 4$).

Step 1: Append 4 zeros: $M' = 11010110110000$

Step 2: Modulo-2 divide 11010110110000 by 10011:

```
11010110110000
```

```
10011
```

```
-----
```

```
10011
```

```
10011
```

```
-----
```

```
00001101
```

```
10011
```

```
-----
```

```
11100
```

```
10011
```

```
-----
```

```
1111  <-- Remainder (FCS)
```

(Full long division omitted for space; remainder = 1110)

Step 3: Transmitted frame $T = 1101011011 + 1110 = 11010110111110$

Receiver divides 11010110111110 by 10011; remainder = 0 $\Rightarrow$ no error.

Standard CRC polynomials:

Table 3.1: Common CRC Standards

Standard	Polynomial	Use
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-16	$x^{16} + x^{15} + x^2 + 1$	USB, HDLC, Modbus
CRC-CCITT	$x^{16} + x^{12} + x^5 + 1$	X.25, HDLC, SD card
CRC-32	$x^{32} + x^{26} + \dots + 1$	Ethernet, ZIP, PNG

What CRC can detect:

- All single-bit errors.
- All double-bit errors (if $G(x)$ has at least three 1s).
- All odd number of bit errors (if $G(x)$ has factor $x + 1$).
- All burst errors of length $\leq r$ (degree of G).
- Most burst errors of length $> r$ (probability of missing = 2^{-r}).

3.3 Checksum

Key Concept: Checksum

Checksum divides data into segments of equal size, adds them together using **one's complement arithmetic**, and transmits the *complement of the sum* (the checksum). The receiver adds all received segments including the checksum; if the result is all 1s (one's complement zero), no error is detected.

Algorithm (Sender):

1. Divide the message into k -bit segments.
2. Add all segments using one's complement addition (wrap-around carry).
3. Take the one's complement of the final sum: this is the **checksum**.
4. Append the checksum to the data and transmit.

Algorithm (Receiver):

1. Add all received segments (including checksum) using one's complement addition.
2. If result is all 1s: **no error detected**.
3. If result is not all 1s: **error detected**.

Worked Example: Checksum – 8-bit segments

Segments: 10101001 and 00111001.

Sum: $10101001 + 00111001 = 11100010$ (no carry).

One's complement (checksum): $\overline{11100010} = 00011101$.

Receiver adds: $10101001 + 00111001 + 00011101$:

$= 11100010 + 00011101 = 11111111$ (all 1s) $\Rightarrow$ no error detected.

Used in: UDP, TCP, IP headers, ICMP.

Table 3.2: Error Detection Methods Comparison

Property	Parity	CRC	Checksum
Overhead	1 bit per word	r bits (e.g., 16 or 32)	k bits (same as segment)
Single-bit error	Yes	Yes	Yes
Burst error detection	Poor	Excellent	Moderate
Error correction	Only with 2D	No	No
Implementation	Very simple	Moderate (hardware)	Simple (software)
Used in	Older systems	Ethernet, HDLC, USB	TCP, UDP, IP

4 Error Correction: Hamming Code

Key Concept: Hamming Code

Hamming Code is a *forward error correction* (FEC) technique that adds redundancy bits (*check bits* or *parity bits*) at specific positions in the data so that the receiver can both **detect** and **correct** single-bit errors *without* retransmission.

4.1 Hamming Distance

Definition: Hamming Distance

The **Hamming distance** $d(c_1, c_2)$ between two codewords is the number of bit positions in which they differ.

$$d = (\text{number of 1s in } c_1 \oplus c_2)$$

For a code with minimum Hamming distance $d_{\min}$:

- To **detect** e errors: $d_{\min} \geq e + 1$
- To **correct** e errors: $d_{\min} \geq 2e + 1$

Hamming code has $d_{\min} = 3$: detects 2-bit errors, corrects 1-bit errors.

4.2 Check Bit Positions

In Hamming code, **check bits** are placed at positions that are **powers of 2**: positions 1, 2, 4, 8, 16, ...

Formula: Number of Check Bits

For m data bits, the number of check bits r satisfies:

$$2^r \geq m + r + 1$$

Example: For $m = 4$ data bits: $2^3 = 8 \geq 4 + 3 + 1 = 8$ ✓, so $r = 3$ check bits. Total codeword length = $m + r = 7$ bits.

4.3 Encoding: Construction of Hamming Code

Step-by-step for a 4-bit data word $d_7d_6d_5d_3$ (data bits at positions 3, 5, 6, 7; check bits at positions 1, 2, 4):

1. Place data bits at non-power-of-2 positions and check bits at power-of-2 positions:

$$\underbrace{p_1}_1 \underbrace{p_2}_2 \underbrace{d_1}_3 \underbrace{p_4}_4 \underbrace{d_2}_5 \underbrace{d_3}_6 \underbrace{d_4}_7$$

2. Compute each check bit using **even parity** over specific bit positions:
 - p_1 (position 1): covers positions 1, 3, 5, 7 (those with bit 0 set in binary)
 - p_2 (position 2): covers positions 2, 3, 6, 7 (those with bit 1 set)
 - p_4 (position 4): covers positions 4, 5, 6, 7 (those with bit 2 set)

Each check bit is set so that the XOR of all covered positions (including the check bit itself) equals 0 (even parity).

Worked Example: Hamming Code for 1011

Data bits: $d_1 = 1, d_2 = 0, d_3 = 1, d_4 = 1$.

Positions: $p_1(1), p_2(2), d_1(3), p_4(4), d_2(5), d_3(6), d_4(7)$

So: $_ _ 1 _ 0 1 1$

Compute check bits (even parity):

$$p_1 : \text{pos } 1, 3, 5, 7 \Rightarrow p_1 \oplus 1 \oplus 0 \oplus 1 = 0 \Rightarrow p_1 = 0$$

$$p_2 : \text{pos } 2, 3, 6, 7 \Rightarrow p_2 \oplus 1 \oplus 1 \oplus 1 = 0 \Rightarrow p_2 = 1$$

$$p_4 : \text{pos } 4, 5, 6, 7 \Rightarrow p_4 \oplus 0 \oplus 1 \oplus 1 = 0 \Rightarrow p_4 = 0$$

Codeword: $p_1 p_2 d_1 p_4 d_2 d_3 d_4 = 0 1 1 0 0 1 1$

4.4 Decoding: Error Detection and Correction

1. Compute the **syndrome**: recalculate each check bit from the received codeword and XOR with the received check bit.
2. Concatenate the syndrome bits (from highest to lowest): this gives the **binary address of the erroneous bit**.
3. If syndrome = 0: no error.
4. If syndrome $\neq 0$: **flip the bit** at the position indicated by the syndrome value.

Worked Example: Error Correction

Sent codeword: 0110011. Received: 0110011 (position 2 corrupted to ... suppose received = 0010011).

Recompute syndromes:

$$s_1 = p_1 \oplus d_1 \oplus d_2 \oplus d_4 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

$$s_2 = p_2 \oplus d_1 \oplus d_3 \oplus d_4 = 0 \oplus 1 \oplus 1 \oplus 1 = 1$$

$$s_4 = p_4 \oplus d_2 \oplus d_3 \oplus d_4 = 0 \oplus 0 \oplus 1 \oplus 1 = 0$$

Syndrome = $s_4 s_2 s_1 = 010_2 = 2_{10}$.

Error is in **position 2**. Flip bit 2: corrected codeword = 0110011.

Exam Tip

For MAKAUT: always show the position table, clearly label which positions are check bits, compute each check bit step by step, and demonstrate syndrome calculation for error detection/correction. Hamming code is a high-value topic (often 5–6 marks).

5 Error Control – ARQ Protocols

Definition: Error Control / ARQ

Automatic Repeat reQuest (ARQ) is the mechanism for achieving *reliable* data transfer over an unreliable link. When the receiver detects an error (or a frame is lost), it signals the sender to **retransmit**. ARQ combines error detection with acknowledgements and timers.

Events requiring retransmission:

- **Lost frame:** Frame never arrives (network drop or corruption so severe the frame is undetectable).
- **Damaged frame:** CRC/parity fails; receiver discards the frame.
- **Lost ACK:** Frame arrived correctly but ACK is lost; sender timer expires.

5.1 Stop-and-Wait ARQ

Key Concept: Stop-and-Wait ARQ

Combines **Stop-and-Wait flow control** with error control. The sender sends one frame, starts a timer, and waits for either an ACK or a NAK (or timeout).

Cases:

1. **Normal:** Receiver gets correct frame → sends ACK → sender sends next frame.
2. **Damaged frame:** Receiver detects error → sends NAK (or discards) → sender retransmits on NAK or timeout.
3. **Lost frame:** Sender timer expires → retransmits.
4. **Lost ACK:** Sender timer expires → retransmits (receiver uses sequence number to detect duplicate).

Sequence numbers: 1 bit (0 and 1 alternate) is sufficient.

Window size: $W = 1$ (one outstanding frame at a time).

5.2 Go-Back-N ARQ (GBN)

Key Concept: Go-Back-N ARQ

In **Go-Back-N**, the sender can have up to $W = 2^n - 1$ unacknowledged frames outstanding (window size). If a frame i is found in error or lost, the sender **re-transmits frame i AND ALL subsequent frames** ($i, i + 1, i + 2, \dots$) that had already been sent.

Key parameters:

- Sender window size: $W_s = 2^n - 1$ (for n -bit sequence numbers).
- Receiver window size: $W_r = 1$ (receiver discards out-of-order frames).
- Uses **cumulative ACKs**: ACK n acknowledges all frames up to and including frame $n - 1$.
- Uses **NAK** (negative acknowledgement): receiver sends NAK i when it receives a frame out of sequence.

Why $W_s \leq 2^n - 1$? If $W_s = 2^n$, after sending all 2^n frames and a timeout, the sender cannot distinguish old duplicate frames from new frames (ambiguous sequence numbers).

Efficiency:

$$\eta_{\text{GBN}} = \begin{cases} 1 & \text{if } p = 0 \\ \frac{W(1-p)}{(1-p) + Wp} & \text{when errors occur} \end{cases} \approx \frac{W}{1 + 2a(W)}$$

where p = probability of frame error.

Table 5.1: Go-Back-N – Advantages and Disadvantages

Advantages	Disadvantages
Simple receiver (window size 1)	Retransmits frames unnecessarily
Higher efficiency than Stop-and-Wait	Wastes bandwidth with retransmissions
Widely implemented	Poor performance on high error-rate links

5.3 Selective Reject ARQ (Selective Repeat)

Key Concept: Selective Reject ARQ

In **Selective Reject** (also called *Selective Repeat*), only the **specific erroneous or lost frame** is retransmitted, not all subsequent frames. The receiver **buffers out-of-order frames** and delivers them in order once the missing frame arrives.

Key parameters:

- Sender window size: $W_s \leq 2^{n-1}$.
- Receiver window size: $W_r = W_s$ (must buffer out-of-order frames).
- **Why $W_s \leq 2^{n-1}$?** To prevent overlap: if sender retransmits frame 0 after a timeout, receiver must distinguish it from a new frame 0 in the next window.

Table 5.2: Selective Reject – Advantages and Disadvantages

Advantages	Disadvantages
Most efficient: only retransmit lost frames	Receiver needs large buffer ($W_r = W_s$)
Best performance on high-error links	More complex receiver logic
	Smaller maximum window size (2^{n-1} vs $2^n - 1$)

5.4 ARQ Protocols – Master Comparison

Table 5.3: Comparison of ARQ Protocols

Property	Stop-and-Wait	Go-Back-N	Selective Reject
Sender window	1	$2^n - 1$	2^{n-1}
Receiver window	1	1	2^{n-1}
Seq. numbers needed	2 (1-bit)	2^n	2^n
Retransmission	1 frame	W frames from error	1 frame only
Receiver buffer	1 frame	1 frame	W_r frames
Complexity	Lowest	Medium	Highest
Efficiency	Lowest	Medium	Highest
Best for	Low delay, low error	Moderate delay	High delay, high error

6 HDLC – High-Level Data Link Control

Definition: HDLC

HDLC (High-level Data Link Control) is a *bit-oriented, synchronous* data link layer protocol developed by ISO. It provides a full set of data link services including framing, error detection (CRC), flow control, and error control (sliding window ARQ), supporting both point-to-point and multipoint links.

6.1 HDLC Stations (Node Types)

- **Primary Station:** Controls the link; sends commands; manages link setup and error recovery.
- **Secondary Station:** Responds to commands from the primary; sends responses.
- **Combined Station:** Can act as both primary and secondary simultaneously (used in Asynchronous Balanced Mode).

6.2 HDLC Transfer Modes

Table 6.1: HDLC Transfer Modes

Mode	Description	Topology
NRM (Normal Response Mode)	Secondary transmits only when polled by Primary	Unbalanced (P/S)
ABM (Asynchronous Balanced Mode)	Combined stations; either may initiate transmission	Balanced (P-P)
ARM (Asynchronous Response Mode)	Secondary initiates without waiting for poll; rare	Unbalanced

6.3 HDLC Frame Structure



Table 6.2: HDLC Frame Fields

Field	Size	Description
Flag	8 bits	01111110: marks start/end of frame; bit stuffing prevents this pattern in data
Address	8+ bits	Address of secondary station
Control	8 or 16 bits	Frame type (I/S/U), sequence numbers, P/F bit
Data	Variable	Upper-layer PDU (absent in some S and U frames)
FCS	16 or 32 bits	CRC-CCITT error detection

6.4 HDLC Frame Types

Table 6.3: HDLC Frame Types

Type	Name	Purpose
I-frame	Information	Carries user data; contains $N(S)$ (send seq#) and $N(R)$ (ACK seq#)
S-frame	Supervisory	Flow/error control (RR, RNR, REJ, SREJ); no data
U-frame	Unnumbered	Link management (SABM, DISC, UA, DM, FRMR); mode setting

S-frame sub-types:

- **RR** (Receive Ready): ACK; ready to receive next frame.
- **RNR** (Receive Not Ready): ACK but sender must pause (flow control).
- **REJ** (Reject): NAK for Go-Back-N; retransmit from $N(R)$.
- **SREJ** (Selective Reject): NAK for a single specific frame.

7 Multiple Access Protocols

Definition: Multiple Access Protocol

A **multiple access protocol** determines how multiple nodes share a single broadcast (shared) communication channel. Without coordination, simultaneous transmissions by multiple nodes result in *collisions*, corrupting all transmitted data.

Classification:

1. **Random Access** (contention-based): ALOHA, CSMA, CSMA/CD, CSMA/CA
2. **Controlled Access** (collision-free): Reservation, Polling, Token Passing
3. **Channelization**: FDMA, TDMA, CDMA

7.1 ALOHA Protocols

7.1.1 Pure ALOHA

Key Concept: Pure ALOHA

In **Pure ALOHA**, any station can transmit *at any time* whenever it has data. If a collision occurs (two frames overlap in time), both are destroyed and each sender waits a random backoff time before retransmitting.

Vulnerability period: A frame of duration T_0 is vulnerable to collision if another frame begins transmission during the interval $[t - T_0, t + T_0]$, a window of $2T_0$.

Formula: Pure ALOHA Throughput

$$S = Ge^{-2G}$$

Maximum throughput at $G = 0.5$:

$$S_{\max} = \frac{1}{2e} \approx 18.4\%$$

where G = offered load (frames per frame-time), S = throughput.

7.1.2 Slotted ALOHA**Key Concept: Slotted ALOHA**

Slotted ALOHA divides time into discrete *slots* of duration T_0 . Stations may only begin transmission at the *start of a slot*. This halves the vulnerability period to T_0 (from $2T_0$ in Pure ALOHA), doubling the maximum throughput.

Formula: Slotted ALOHA Throughput

$$S = Ge^{-G}$$

Maximum throughput at $G = 1$:

$$S_{\max} = \frac{1}{e} \approx 36.8\%$$

Table 7.1: Pure ALOHA vs. Slotted ALOHA

Property	Pure ALOHA	Slotted ALOHA
Transmission time	Any time	Start of a time slot only
Vulnerability period	$2T_0$	T_0
Max throughput	18.4% ($G = 0.5$)	36.8% ($G = 1$)
Synchronization	Not required	Requires global clock sync
Complexity	Very simple	Moderate

7.2 CSMA (Carrier Sense Multiple Access)**Key Concept: CSMA**

CSMA improves on ALOHA by requiring each station to *listen to the channel* (carrier sense) before transmitting. If the channel is busy, the station defers. This reduces but does not eliminate collisions (due to propagation delay: two stations may both sense the channel as idle and begin transmitting simultaneously).

CSMA Persistence strategies:

- **1-persistent:** If channel idle, transmit immediately. If busy, wait until idle then transmit. (Aggressive – high collision risk.)
- **Non-persistent:** If busy, wait a *random* time before sensing again. (Lower collisions, higher delay, lower efficiency.)
- **p-persistent:** If idle, transmit with probability p , or defer by one slot with probability $1 - p$. (Compromise.)

7.2.1 CSMA/CD (Collision Detection)

Key Concept: CSMA/CD

CSMA/CD (used in **IEEE 802.3 Ethernet**) adds *collision detection*: while transmitting, each station monitors the channel. If a collision is detected (the received signal differs from the transmitted signal), all transmitting stations immediately abort, send a jam signal, wait a random backoff time, and retry.

Binary Exponential Backoff: After the k -th collision, wait a random number of slot times chosen from $[0, 2^k - 1]$. After 16 collisions, report failure.

Minimum frame size: For CSMA/CD to work, the frame transmission time must be $\geq 2 \times$ propagation delay (round-trip time):

$$T_f \geq 2T_p \quad \Rightarrow \quad L_{\min} = 2 \times T_p \times R$$

For 10 Mbps Ethernet (500 m max, $T_p = 2.5 \mu\text{s}$ round trip): $L_{\min} = 64$ bytes.

7.2.2 CSMA/CA (Collision Avoidance)

Used in **IEEE 802.11 Wi-Fi** where collision detection is impractical (hidden node problem, half-duplex radio). Stations use **ACKs**, **random backoff (DIFS/SIFS)**, and optionally **RTS/CTS** to *avoid* collisions rather than detect them.

7.3 Collision-Free Protocols

7.3.1 Reservation Protocol

A frame is divided into a *reservation mini-slot* and a *data slot*. Stations announce reservations in mini-slots; data frames are sent in reserved slots without collision.

7.3.2 Polling (Roll-call Polling)

Key Concept: Polling

A **primary (master) station** polls each secondary station in turn, granting it permission to transmit. No collisions are possible.

Variants:

- **Roll-call polling:** Primary cycles through all stations.
- **Hub polling (Daisy chain):** Permission passes sequentially.

7.3.3 Token Passing

Key Concept: Token Passing

A special control frame called the **token** circulates among stations. Only the station *holding the token* may transmit. After transmission (or timeout), the token is passed to the next station.

- **Token Bus** (IEEE 802.4): Logical ring over physical bus.
- **Token Ring** (IEEE 802.5): Logical and physical ring.
- **FDDI**: Token ring over fiber at 100 Mbps with dual ring.

Advantages: Deterministic access, no collisions, predictable performance.

Disadvantages: Token management overhead; failure of one node can disrupt the ring.

8 IEEE Standards for LAN

Definition: IEEE 802 Standards

The **IEEE 802** project (1980) defines standards for local area networks and metropolitan area networks. The standards cover the Physical Layer and the Data Link Layer (split into MAC and LLC sub-layers).

Table 8.1: IEEE 802 Standard Family

Standard	Description
802.1	LAN Architecture, bridging, VLANs, spanning tree
802.2	LLC (Logical Link Control) – common to all 802 standards
802.3	CSMA/CD (Ethernet)
802.4	Token Bus
802.5	Token Ring
802.6	MAN (DQDB)
802.11	Wireless LAN (Wi-Fi)
802.15	Wireless PAN (Bluetooth)
802.16	Wireless MAN (WiMAX)

8.1 IEEE 802.3 – Ethernet (CSMA/CD)

Key Concept: IEEE 802.3

IEEE 802.3 defines the Ethernet standard, using **CSMA/CD** as the medium access method. It is the dominant wired LAN technology worldwide.

Ethernet Frame Format:

Preamble	SFD	Dest. MAC	Src. MAC	Length/Type	Data (Payload)
7 bytes	1 byte	6 bytes	6 bytes	2 bytes	46–1500 bytes
FCS (4 bytes – CRC-32)					

Key field descriptions:

- **Preamble:** 7 bytes of 10101010... for clock synchronization.
- **SFD (Start Frame Delimiter):** 10101011 – marks start of frame.
- **Dest./Src. MAC:** 6-byte (48-bit) hardware addresses.
- **Length/Type:** Frame length (802.3) or EtherType (Ethernet II).
- **Data:** 46–1500 bytes payload (padded to 46 if shorter).
- **FCS:** 4-byte CRC-32 for error detection.

Ethernet Variants:

Table 8.2: IEEE 802.3 Variants

Name	Speed	Medium	Max Segment
10BASE5	10 Mbps	Thicknet coax	500 m
10BASE2	10 Mbps	Thinnet coax	185 m
10BASE-T	10 Mbps	UTP Cat 3	100 m
100BASE-TX	100 Mbps	UTP Cat 5	100 m
1000BASE-T	1 Gbps	UTP Cat 5e	100 m
10GBASE-T	10 Gbps	UTP Cat 6a	100 m
1000BASE-SX	1 Gbps	Multimode fiber	550 m

Ethernet addressing:

- **Unicast:** LSB of first byte = 0. Sent to one device.
- **Multicast:** LSB of first byte = 1. Sent to a group.
- **Broadcast:** All 48 bits = 1 (FF:FF:FF:FF:FF:FF). Sent to all.

8.2 IEEE 802.4 – Token Bus

Key Concept: IEEE 802.4 – Token Bus

IEEE 802.4 Token Bus uses a **physical bus** topology but implements a **logical ring** for token passing. Stations are assigned logical addresses; the token passes in order of logical address, not physical position.

Features:

- Deterministic access: bounded maximum wait time (unlike CSMA/CD).
- Supports **priority levels** (four priority classes: 0, 2, 4, 6).
- Used in **industrial automation** (MAP – Manufacturing Automation Protocol).
- Frame format similar to token ring with additional control fields.
- **Token management:** handles lost tokens, duplicate tokens, new station joining/leaving the ring.

Advantages: Real-time guarantees, priority support, deterministic.

Disadvantages: Complex token management, largely replaced by Fast Ethernet in modern industry.

8.3 IEEE 802.5 – Token Ring

Key Concept: IEEE 802.5 – Token Ring

IEEE 802.5 Token Ring uses a physical **ring topology** with **token passing** as the MAC method. Data travels in one direction around the ring. A station can transmit only when it holds the token.

Operation:

1. A free token circulates around the ring.
2. A station wishing to transmit *captures* the token (changes the token to a “busy” frame) and appends its data.
3. The frame travels around the ring; each station reads the destination address and *copies* the frame if it is the intended recipient.
4. The frame returns to the originating station, which removes it from the ring and releases a new free token.

Token Ring Frame Format:

[SD | AC | FC | Dest. | Src. | Data | FCS | ED | FS]

- **SD/ED:** Start/End delimiter.
- **AC (Access Control):** Contains the token bit (T), priority (PPP), reservation (RRR), and monitor (M) bits.
- **FC (Frame Control):** Distinguishes data frames from MAC frames.
- **FCS:** CRC-32 error detection.
- **FS (Frame Status):** Address Recognized (A) and Frame Copied (C) bits set by recipient.

Encoding: Differential Manchester encoding (IEEE 802.5).

Speed: 4 Mbps or 16 Mbps.

Wiring: Star-wired ring using MAU (Medium Attachment Unit) / MSAU (Multi-Station Access Unit).

Table 8.3: Comparison of IEEE 802.3, 802.4, and 802.5

Property	IEEE 802.3 (Ethernet)	IEEE 802.4 (Token Bus)	IEEE 802.5 (Token Ring)
Topology	Bus / Star	Physical bus, logical ring	Physical ring (star-wired)
MAC method	CSMA/CD	Token passing	Token passing
Deterministic	No (collisions)	Yes	Yes
Priority	No (basic)	Yes (4 levels)	Yes (8 levels)
Encoding	Manchester / 4B5B	Various	Differential Manchester
Speed	10 Mbps – 10+ Gbps	1/5/10 Mbps	4 / 16 Mbps
Application	General LAN	Industrial automation	Older corporate LAN
Collision	Yes (CSMA/CD)	None	None
Standard status	Current, dominant	Mostly obsolete	Largely replaced

9 Numerical Problems – Module 3

Worked Numericals

Problem 1 (CRC): Message = 101100, Generator = 1011.

Append $r = 3$ zeros: 101100000. Divide by 1011:

$101100000 \div 1011$: quotient steps via XOR...

Remainder = 100

Transmitted frame = 101100 + 100 = 101100100.

Problem 2 (Hamming): Data = 1001. Find r : $2^3 = 8 \geq 4 + 3 + 1 = 8 \checkmark$, $r = 3$.

Positions: $p_1, p_2, d_1 = 1, p_4, d_2 = 0, d_3 = 0, d_4 = 1$.

p_1 (pos 1,3,5,7): $p_1 \oplus 1 \oplus 0 \oplus 1 = 0 \Rightarrow p_1 = 0$

p_2 (pos 2,3,6,7): $p_2 \oplus 1 \oplus 0 \oplus 1 = 0 \Rightarrow p_2 = 0$

p_4 (pos 4,5,6,7): $p_4 \oplus 0 \oplus 0 \oplus 1 = 0 \Rightarrow p_4 = 1$

Codeword: **0011001**.

Problem 3 (Sliding Window efficiency): $T_p = 200$ ms, $T_f = 1$ ms, $W = 127$ frames ($n = 7$, GBN).

$a = 200/1 = 200$, $1 + 2a = 401$.

$W = 127 < 401 \Rightarrow \eta = 127/401 \approx 31.7\%$.

Problem 4 (Pure vs Slotted ALOHA): At $G = 0.5$: $S_{\text{Pure}} = 0.5e^{-1} \approx 18.4\%$.

At $G = 1$: $S_{\text{Slotted}} = 1 \cdot e^{-1} \approx 36.8\%$.

10 Practice Questions (MAKAUT Style)

Short Answer Questions (2–3 marks)

- Q1.** What are the functions of the Data Link Layer?
- Q2.** Define framing. Name four framing methods.
- Q3.** What is flow control? Why is it needed?
- Q4.** Derive the efficiency formula for Stop-and-Wait protocol.
- Q5.** State two differences between Go-Back-N and Selective Reject ARQ.
- Q6.** What is parity check? Distinguish between simple and 2D parity.
- Q7.** What is CRC? What errors can CRC detect?
- Q8.** Define Hamming distance. What is the minimum Hamming distance for single-bit error correction?
- Q9.** What is HDLC? Name the three types of HDLC frames.
- Q10.** What is the maximum throughput of Pure ALOHA and Slotted ALOHA?
- Q11.** Distinguish between CSMA/CD and CSMA/CA.
- Q12.** What is token passing? In which IEEE standard is it used?

Long Answer Questions (5–6 marks)

- Q1.** Explain the CRC algorithm with a suitable example. What are the standard CRC polynomials used?
- Q2.** For data bits 1011, generate the Hamming code. If bit 5 of the transmitted codeword is corrupted, show how the error is detected and corrected.
- Q3.** Explain Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Reject ARQ. Compare them with respect to window size, receiver buffer, and efficiency.
- Q4.** Explain the ALOHA protocol (Pure and Slotted). Derive the throughput formula for each and find the maximum throughput.
- Q5.** Compare IEEE 802.3, IEEE 802.4, and IEEE 802.5 with respect to topology, MAC method, frame format, and applications.
- Q6.** Explain the HDLC protocol. Describe its frame format, frame types, and transfer modes.

11 Module 3 – Key Takeaways

Summary Checklist

- ✓ DLL functions: framing, addressing, error detection/correction, flow control, access control.
- ✓ Stop-and-Wait: $W = 1$, $\eta = 1/(1 + 2a)$. Sliding window: $\eta = W/(1 + 2a)$ if $W < 1 + 2a$.
- ✓ Parity: detects odd-number errors. CRC: powerful, detects burst errors $\leq r$. Checksum: TCP/IP.
- ✓ Hamming: $2^r \geq m + r + 1$; check bits at powers of 2; syndrome gives error position.
- ✓ GBN: sender $W \leq 2^n - 1$, receiver $W_r = 1$, retransmits all from error.
- ✓ Selective Repeat: $W_s \leq 2^{n-1}$, $W_r = W_s$, retransmits only lost frame.
- ✓ HDLC: I-frame (data), S-frame (flow/error control), U-frame (link management).
- ✓ Pure ALOHA: $S_{\max} = 18.4\%$ at $G = 0.5$. Slotted ALOHA: $S_{\max} = 36.8\%$ at $G = 1$.
- ✓ IEEE 802.3: Ethernet, CSMA/CD, dominant. IEEE 802.4: Token Bus, industrial. IEEE 802.5: Token Ring, Differential Manchester.

Module 4 — The Network Layer

Routing Algorithms, IP Addressing, RIP, OSPF, BGP, NAT, ICMP

10 Hours · 12 Marks

1 Network Layer – Overview

Definition: Network Layer

The **Network Layer** (OSI Layer 3) is responsible for *end-to-end* logical addressing and *routing* of packets from a source host to a destination host across multiple interconnected networks (internetworking). It hides the complexity of the underlying physical networks.

Key functions:

1. **Logical Addressing (IP):** Assigns globally unique IP addresses to hosts.
2. **Routing:** Selects the best path from source to destination.
3. **Packet Forwarding:** Moves a packet from input to appropriate output port.
4. **Fragmentation and Reassembly:** Breaks large packets to fit MTU; reassembles at destination.
5. **Error Reporting:** Uses ICMP to report delivery problems.
6. **Congestion Control:** Limits traffic to prevent network overload.

2 Routing Algorithms

Definition: Routing Algorithm

A **routing algorithm** determines the *optimal path* (or set of paths) through a network from source to destination. The result is stored in a **routing table** at each router.

Classification of routing algorithms:

- **Adaptive (Dynamic):** Routes change based on current network conditions (traffic, topology).
- **Non-adaptive (Static):** Routes are pre-computed and fixed; do not respond to changes.
- **Centralized vs. Distributed:** Whether one node or all nodes compute routes.
- **Global (Link State) vs. Decentralized (Distance Vector).**

2.1 Shortest Path Algorithm (Dijkstra's)

Key Concept: Dijkstra's Shortest Path

Dijkstra's algorithm computes the shortest (least-cost) path from a *single source* to *all other nodes* in a weighted graph. It is the basis of **Link State Routing** (OSPF).

Algorithm:

1. Initialize: distance to source = 0; all others = ∞ . Mark all unvisited.
2. Select the unvisited node u with the smallest known distance.
3. For each unvisited neighbor v of u :

$$d(v) = \min(d(v), d(u) + w(u, v))$$

Update $d(v)$ and predecessor if shorter path found.

4. Mark u as visited. Repeat until all nodes visited.
5. Result: shortest path tree rooted at source.

Complexity: $O(n^2)$ with simple implementation; $O((n + e) \log n)$ with a min-heap.

Properties: Finds globally optimal paths; requires complete knowledge of network topology (link state).

Worked Example: Dijkstra Example

Consider nodes A, B, C, D, E with costs:

A-B=2, A-C=5, B-C=1, B-D=7, C-E=3, D-E=1.

Source = A. Steps:

- Init: A=0, B= ∞ , C= ∞ , D= ∞ , E= ∞
- Visit A: B=2, C=5
- Visit B (dist=2): C= $\min(5, 2+1)=3$, D= $2+7=9$
- Visit C (dist=3): E= $3+3=6$
- Visit E (dist=6): D= $\min(9, 6+1)=7$
- Visit D (dist=7): done

Shortest paths from A: B=2, C=3, D=7, E=6.

2.2 Flow-Based Routing

Key Concept: Flow-Based Routing

Flow-based routing determines routes by minimizing average packet delay across the entire network, considering both topology *and traffic flow* (load). It computes routes that balance traffic to avoid congestion hot-spots.

The average delay on a link carrying λ_i packets/sec with capacity μ_i packets/sec (using M/M/1 queuing model):

$$T_i = \frac{1}{\mu_i - \lambda_i}$$

Total average delay: $T = \frac{1}{\gamma} \sum_i \lambda_i T_i$ where γ = total input traffic.

Routes are chosen to minimize T subject to flow conservation constraints.

2.3 Distance Vector Routing (Bellman-Ford)

Key Concept: Distance Vector Routing

Each router maintains a **distance vector** table: the estimated cost to every other router via each neighbor. Routers periodically share their distance vectors with *immediate neighbors only*. Routes are computed iteratively using the **Bellman-Ford equation**:

$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

where $c(x, v)$ is the cost to neighbor v , and $D_v(y)$ is v 's estimated distance to y .

Operation:

1. Each router initializes its table with direct link costs.
2. Periodically broadcasts its distance vector to neighbors.
3. On receiving a neighbor's vector, updates its own table using Bellman-Ford.
4. Converges when no more updates occur.

Count-to-Infinity Problem: When a link fails, routers may keep incrementing the cost in a loop, never converging. Solutions:

- **Split horizon:** Do not advertise a route back to the neighbor from which it was learned.
- **Poison reverse:** Advertise routes back with infinite cost (∞).
- **Hold-down timer:** Ignore updates for a route for a period after it goes down.

Used in: RIP (Routing Information Protocol).

Table 2.1: Link State vs. Distance Vector Comparison

Property	Link State (Dijkstra)	Distance Vector (Bellman-Ford)
Knowledge required	Full network topology	Only neighbor distances
Information shared	Link state to ALL nodes (flooding)	Distance vector to neighbors only
Convergence speed	Fast	Slow (count-to-infinity)
Computation	Done locally at each node	Distributed, iterative
Bandwidth usage	High (initial flooding)	Low (periodic neighbor updates)
Scalability	Better (hierarchical OSPF)	Poor for large networks
Examples	OSPF, IS-IS	RIP, BGP (path vector variant)

2.4 Broadcast Routing

Definition: Broadcast Routing

Broadcast routing delivers a packet from a source to *all* nodes in the network. Naive flooding (forwarding on all links except arrival) works but creates duplicate packets.

Efficient broadcast methods:

- **Reverse Path Forwarding (RPF):** A router forwards a broadcast packet only if it arrives on the link that is on the shortest path back to the source. Eliminates most duplicates without a spanning tree.
- **Spanning Tree Broadcast:** Build a spanning tree of the network; broadcast only along tree edges. Zero duplicate packets but requires tree construction and maintenance.
- **Sequence-number controlled flooding:** Each packet has a source address and sequence number; routers discard duplicates.

2.5 Multicast Routing

Definition: Multicast Routing

Multicast routing delivers a packet from one source to a *specific group* of receivers (neither unicast nor broadcast). Receivers *join* a multicast group (identified by a Class D IP address: 224.0.0.0 – 239.255.255.255).

Multicast routing approaches:

- **DVMRP** (Distance Vector Multicast Routing Protocol): Uses RPF + pruning. Build a delivery tree; prune branches with no receivers.
- **PIM-SM** (Protocol-Independent Multicast – Sparse Mode): Uses a shared Rendezvous Point (RP); suitable when receivers are sparse.
- **IGMP** (Internet Group Management Protocol): Used by hosts to join/leave multicast groups; routers use IGMP to track group membership.

3 IP Addressing

3.1 IPv4 Addressing

Definition: IPv4 Address

An **IPv4 address** is a 32-bit logical identifier assigned to each network interface. Written in *dotted-decimal notation*: four 8-bit octets separated by dots (e.g., 192.168.1.10).

IPv4 Address Classes (Classful, legacy):

Table 3.1: IPv4 Address Classes

Class	First Bits	Range (1st octet)	Default Mask	Hosts per Network
A	0xxx	1 – 126	/8 (255.0.0.0)	$2^{24} - 2 = 16,777,214$
B	10xx	128 – 191	/16 (255.255.0.0)	$2^{16} - 2 = 65,534$
C	110x	192 – 223	/24 (255.255.255.0)	$2^8 - 2 = 254$
D	1110	224 – 239	N/A	Multicast group
E	1111	240 – 255	N/A	Reserved/Experimental

Special IPv4 addresses:

- 127.0.0.1: Loopback address (localhost).
- 0.0.0.0: Unspecified address (this host on this network).
- 255.255.255.255: Limited broadcast (all hosts on local network).
- Host bits all 0: Network address (identifies the network).
- Host bits all 1: Directed broadcast.

Private (RFC 1918) address ranges:

- Class A: 10.0.0.0/8
- Class B: 172.16.0.0/12 (172.16.0.0 – 172.31.255.255)
- Class C: 192.168.0.0/16

3.1.1 IPv4 Packet Header

Table 3.2: IPv4 Header Fields

Field	Size	Description
Version	4 bits	IP version (4 for IPv4)
IHL	4 bits	Internet Header Length in 32-bit words (min 5 = 20 bytes)
Type of Service	8 bits	DSCP/ToS; QoS differentiation
Total Length	16 bits	Total packet size (header + data) in bytes; max 65,535
Identification	16 bits	Fragment identification number
Flags	3 bits	DF (Don't Fragment), MF (More Fragments)
Fragment Offset	13 bits	Offset of this fragment in the original datagram (in 8-byte units)
TTL	8 bits	Time to Live; decremented by each router; packet discarded at 0
Protocol	8 bits	Upper-layer protocol: TCP=6, UDP=17, ICMP=1
Header Checksum	16 bits	Error check on header only (not data)
Source IP	32 bits	Sender's IP address
Destination IP	32 bits	Receiver's IP address
Options	Variable	Security, timestamp, source routing (optional)

3.2 Subnetting and CIDR

3.2.1 Subnetting

Key Concept: Subnetting

Subnetting divides a single large network (IP block) into multiple smaller *sub-networks* (*subnets*) by borrowing bits from the host portion of the address. This improves security, reduces broadcast domains, and makes better use of the IP address space.

Formula: Subnet Calculations

Given a subnet mask with n network bits and $h = 32 - n$ host bits:

$$\text{Number of subnets} = 2^s \quad (\text{where } s = \text{bits borrowed from host})$$

$$\text{Hosts per subnet} = 2^h - 2$$

(subtract 2: one for network address, one for broadcast)

Subnet mask in binary: n ones followed by h zeros.

Worked Example: Subnetting Example

Network: 192.168.10.0/24. Divide into 4 equal subnets.

Borrow 2 bits from host: new mask = /26 = 255.255.255.192.

Hosts per subnet = $2^6 - 2 = 62$.

Subnet 1:	192.168.10.0/26	hosts .1 – .62, broadcast .63
Subnet 2:	192.168.10.64/26	hosts .65 – .126, broadcast .127
Subnet 3:	192.168.10.128/26	hosts .129 – .190, broadcast .191
Subnet 4:	192.168.10.192/26	hosts .193 – .254, broadcast .255

3.2.2 CIDR (Classless Inter-Domain Routing)**Key Concept: CIDR**

CIDR (RFC 4632) eliminates the rigid class-based addressing system. An IP address is written with a *prefix length* (e.g., 172.16.0.0/20) that explicitly specifies how many bits are the network portion. CIDR allows:

- **Variable-Length Subnet Masking (VLSM):** Different subnets of different sizes within the same organization.
- **Route aggregation (supernetting):** Multiple contiguous networks combined into one routing entry, reducing routing table size.

Worked Example: CIDR Route Aggregation

Routes: 200.23.16.0/23, 200.23.18.0/23, 200.23.20.0/23, 200.23.22.0/23 can be aggregated to 200.23.16.0/21 (common prefix of 21 bits). One entry replaces four.

3.3 IPv6 Addressing**Key Concept: IPv6**

IPv6 uses **128-bit** addresses (versus 32-bit IPv4), providing $2^{128} \approx 3.4 \times 10^{38}$ unique addresses – effectively unlimited. It was designed to solve IPv4 address exhaustion and includes improvements in routing, security (IPsec), and QoS.

IPv6 address notation:

- 128 bits written as **8 groups of 4 hex digits** separated by colons: 2001:0db8:85a3:0000:0000:8b00:0000:0000
- **Abbreviation rules:**
 - Leading zeros in each group can be omitted: 0db8 → db8
 - One (longest) run of consecutive all-zero groups replaced by :: 2001:db8:85a3::8a2e:370:1

IPv6 Address Types:

- **Unicast:** One-to-one. Types: Global (2000::/3), Link-local (fe80::/10), Loop-back (:::1).
- **Multicast:** One-to-many (starts with ff00::/8). *No broadcast in IPv6* – replaced by multicast.

- **Anycast:** One-to-nearest; same address assigned to multiple interfaces; packet delivered to nearest one.

IPv6 Header improvements vs. IPv4:

Table 3.3: IPv4 vs. IPv6 Header Comparison

Property	IPv4	IPv6
Address size	32 bits	128 bits
Header size	Variable (20–60 bytes)	Fixed 40 bytes
Checksum	Yes (header only)	Removed (left to transport)
Fragmentation	By routers and hosts	By source host only
Options	In header (variable)	Extension headers (optional)
IPsec	Optional	Mandatory support
Broadcast	Yes	No (replaced by multicast)
Auto-configuration	DHCP only	SLAAC (Stateless Address AutoConfig)

4 Routing Protocols

4.1 RIP – Routing Information Protocol

Key Concept: RIP

RIP is a **Distance Vector** routing protocol that uses *hop count* as the metric. Maximum hop count = **15** (16 = infinity = unreachable). It is an **Interior Gateway Protocol (IGP)** used within an Autonomous System (AS).

RIP Operation:

- Routers broadcast their full routing table every **30 seconds**.
- Uses **Bellman-Ford** algorithm.
- **RIPv1:** Classful, no subnet mask in updates, broadcast updates.
- **RIPv2:** Classless (CIDR support), sends subnet masks, multicast updates (224.0.0.9).
- **RIPng:** IPv6 version.

Timers:

- **Update timer:** 30 seconds (send full table to neighbors).
- **Invalid timer:** 180 seconds (route marked invalid if no update received).
- **Hold-down timer:** 180 seconds (ignore better routes after failure to prevent loops).
- **Flush timer:** 240 seconds (remove route from table).

Table 4.1: RIP – Advantages and Disadvantages

Advantages	Disadvantages
Simple, easy to configure	Max 15 hops – not scalable
Widely supported	Slow convergence (count-to-infinity)
Low CPU/memory overhead	Metric = hop count only (ignores bandwidth)
	Sends full table every 30s (bandwidth waste)

4.2 OSPF – Open Shortest Path First

Key Concept: OSPF

OSPF is a **Link State** routing protocol using **Dijkstra's algorithm**. Each router builds a complete map (link state database) of the network topology and computes shortest paths independently. OSPF uses **cost** (inversely proportional to bandwidth) as its metric.

OSPF Operation:

1. **Neighbor discovery:** Routers send *Hello* packets to discover neighbors and elect Designated Router (DR) and Backup DR (BDR) on broadcast networks.
2. **Link State Advertisement (LSA):** Each router floods its LSA (describing its links and costs) to *all* other routers in the area.
3. **Link State Database (LSDB):** All routers build an identical LSDB.
4. **SPF Calculation:** Each router runs Dijkstra on its LSDB to compute the shortest path tree.
5. **Routing table:** Built from the shortest path tree.

OSPF Hierarchical Design (Areas):

- Large networks are divided into **areas** to limit LSA flooding.
- **Area 0 (Backbone area):** All other areas must connect to it.
- **ABR (Area Border Router):** Connects an area to Area 0.
- **ASBR (AS Boundary Router):** Connects OSPF domain to external AS.

OSPF Cost metric:

$$\text{Cost} = \frac{10^8}{\text{Bandwidth (bps)}}$$

Example: 100 Mbps link $\rightarrow$ cost = $10^8/10^8 = 1$. 1 Mbps link $\rightarrow$ cost = $10^8/10^6 = 100$.

Table 4.2: RIP vs. OSPF Comparison

Property	RIP	OSPF
Algorithm	Distance Vector (Bellman-Ford)	Link State (Dijkstra)
Metric	Hop count (max 15)	Cost (bandwidth-based)
Convergence	Slow	Fast
Scalability	Small networks only	Large networks (hierarchical areas)
Updates	Full table, every 30s	Triggered LSAs; partial updates
Bandwidth usage	High (periodic full table)	Lower (triggered, incremental)
Authentication	Optional (v2)	Yes (MD5/SHA)
VLSM/CIDR	RIPv2 only	Yes

4.3 BGP – Border Gateway Protocol

Key Concept: BGP

BGP is the *inter-domain* routing protocol of the Internet – the “glue” that holds the Internet together. It is a **Path Vector** protocol that routes between **Autonomous Systems (AS)**. Each AS is identified by a unique **AS Number (ASN)**.

BGP key concepts:

- **eBGP (External BGP):** Between routers in *different* ASes.
- **iBGP (Internal BGP):** Between routers *within* the same AS.
- **Path Vector:** BGP advertises the full AS path to each destination, not just a distance. This prevents routing loops (loop detection by checking if own ASN is in the AS path).
- **BGP attributes:** AS_PATH, NEXT_HOP, LOCAL_PREF, MED, COMMUNITY – used to implement routing policy.
- **BGP uses TCP port 179** for reliable neighbor sessions.
- **Policy-based routing:** BGP allows ISPs to implement business policies (prefer certain paths, avoid certain ASes).

Table 4.3: IGP vs. EGP (BGP) Comparison

Property	IGP (RIP, OSPF)	EGP (BGP)
Scope	Within an AS	Between ASes
Goal	Optimal path (metric-based)	Policy-based routing
Algorithm	Distance Vector / Link State	Path Vector
Scalability	Moderate	Internet-scale (millions of routes)
Transport	UDP (RIP) / IP (OSPF)	TCP port 179

5 NAT – Network Address Translation

Key Concept: NAT

NAT translates *private* (RFC 1918) IP addresses used inside a local network to a *public* IP address before forwarding packets to the Internet. This conserves the limited IPv4 address space and provides a measure of security (internal addresses are hidden).

How NAT works:

1. Host with private IP 192.168.1.5:1234 sends a packet to the Internet.
2. The NAT router replaces the source IP/port with its public IP 203.0.113.1:5001 and records the mapping in a **NAT table**.
3. The remote server replies to 203.0.113.1:5001.
4. The NAT router looks up the table, translates back to 192.168.1.5:1234, and forwards to the internal host.

Types of NAT:

- **Static NAT:** One-to-one mapping (one private ↔ one public).
- **Dynamic NAT:** Pool of public IPs; assigned on demand.
- **PAT / NAPT (Port Address Translation):** Many private IPs mapped to *one* public IP using different port numbers. Most common (home router).

Advantages of NAT:

- Conserves IPv4 addresses.
- Hides internal network topology (security benefit).
- Flexible internal addressing.

Disadvantages of NAT:

- Breaks end-to-end connectivity (violates IP architecture).
- Complications for peer-to-peer applications and VoIP.
- Stateful – NAT table is a single point of failure.
- Not needed in IPv6 (abundant address space).

6 ICMP – Internet Control Message Protocol

Definition: ICMP

ICMP (RFC 792) is a companion protocol to IP (Protocol field = 1) used for *error reporting* and *network diagnostics*. ICMP messages are encapsulated within IP datagrams. ICMP does not make IP reliable; it only *reports* errors back to the source.

Key ICMP message types:

Table 6.1: Important ICMP Message Types

Type	Code	Name	Description
0	0	Echo Reply	Response to ping
3	0–15	Destination Unreachable	Net/Host/Port/Protocol unreachable
4	0	Source Quench	Congestion control (deprecated)
5	0–3	Redirect	Better route exists
8	0	Echo Request	Ping request
11	0–1	Time Exceeded	TTL expired (used by traceroute)
12	0–1	Parameter Problem	Malformed IP header

ICMP Applications:

- **Ping:** Sends Echo Request (type 8); receives Echo Reply (type 0). Tests reachability and round-trip time.
- **Traceroute:** Sends packets with incrementing TTL values (1, 2, 3, ...). Each router that discards a TTL-expired packet sends ICMP Time Exceeded (type 11), revealing the path hop by hop.

7 Numerical Problems – Module 4

Worked Numericals

Problem 1 (Subnetting): IP: 172.16.0.0/16. Create 8 subnets.

Borrow $\lceil \log_2 8 \rceil = 3$ bits. New mask: /19 = 255.255.224.0

Hosts per subnet: $2^{13} - 2 = 8190$.

Subnet 1: 172.16.0.0/19 (hosts .0.1 – .31.254)

Subnet 2: 172.16.32.0/19 (hosts .32.1 – .63.254) etc.

Problem 2 (CIDR): Addresses needed: 1000. CIDR block: $2^{10} = 1024 \geq 1000$.
Use /22.

$$\text{Block size} = 2^{32-22} = 2^{10} = 1024 \text{ addresses}$$

Problem 3 (Dijkstra): Graph: A-B(1), A-C(4), B-C(2), B-D(5), C-D(1).

Source A: init A=0, B= ∞ , C= ∞ , D= ∞ .

Visit A: B=1, C=4. Visit B: C= $\min(4, 1+2)=3$, D= $1+5=6$. Visit C:
D= $\min(6, 3+1)=4$.

Visit D: done. Paths: B=1, C=3, D=4.

8 Practice Questions (MAKAUT Style)**Short Answer (2–3 marks)**

- Q1.** What are the functions of the Network Layer?
- Q2.** What is the count-to-infinity problem? How is it solved?
- Q3.** Differentiate between static and dynamic routing.
- Q4.** What is subnetting? Why is it used?
- Q5.** What is CIDR? How does it help reduce routing table size?
- Q6.** List four differences between IPv4 and IPv6.
- Q7.** What is NAT? State its advantages and disadvantages.
- Q8.** What is ICMP? Name three ICMP message types with their use.
- Q9.** Differentiate between RIP and OSPF.
- Q10.** What is BGP? Why is it called a path vector protocol?

Long Answer (5–6 marks)

- Q1.** Explain Dijkstra's shortest path algorithm with a worked example on a graph of 5 nodes.
- Q2.** Explain Distance Vector Routing. What is the count-to-infinity problem and how is it solved?
- Q3.** Explain IPv4 header format in detail. What is the purpose of each field?
- Q4.** Subnet 192.168.1.0/24 into 6 subnets of varying sizes using VLSM.
- Q5.** Compare RIP, OSPF, and BGP routing protocols with respect to algorithm, metric, scalability, and use case.
- Q6.** Explain NAT with a diagram. Describe PAT (Port Address Translation) in detail.

9 Module 4 – Key Takeaways

Summary Checklist

- ✓ Dijkstra: $d(v) = \min(d(v), d(u) + w(u, v))$; $O(n^2)$; used in OSPF.
- ✓ Distance Vector: Bellman-Ford; count-to-infinity; used in RIP.
- ✓ Subnetting: borrow s bits; 2^s subnets; $2^h - 2$ hosts.
- ✓ CIDR: prefix notation $/n$; route aggregation.
- ✓ IPv4: 32-bit, classes A/B/C/D/E; private: 10/172.16/192.168.
- ✓ IPv6: 128-bit; fixed 40-byte header; no broadcast; SLAAC.
- ✓ RIP: distance vector, hop count ≤ 15 , 30s updates.
- ✓ OSPF: link state, cost=BW-based, areas, fast convergence.
- ✓ BGP: path vector, inter-AS, TCP:179, policy-based.
- ✓ NAT: private-to-public translation; PAT uses port numbers.
- ✓ ICMP: ping (type 8/0), traceroute (type 11), unreachable (type 3).

Module 5 — The Transport Layer

UDP, TCP, Congestion Control, Flow Control, Port Addressing

8 Hours · 10 Marks

1 Transport Layer – Overview and Functions

Definition: Transport Layer

The **Transport Layer** (OSI Layer 4) provides *end-to-end logical communication* between *application processes* running on different hosts. While the Network Layer provides host-to-host delivery, the Transport Layer provides **process-to-process** delivery using **port numbers**.

1.1 Functions of the Transport Layer

1. **Process-to-Process Delivery:** Identifies the specific application process using **port numbers** (0–65535). The combination of IP address + port number is called a **socket**.
2. **Segmentation and Reassembly:** Divides application messages into **segments** (TCP) or **datagrams** (UDP) for transmission. Reassembles at the destination.
3. **Connection Management:** TCP establishes (3-way handshake), maintains, and terminates (4-way handshake) connections. UDP is connectionless.
4. **Reliable Data Transfer (TCP):** Guarantees delivery using acknowledgements, retransmissions (ARQ), and sequence numbers.
5. **Flow Control (TCP):** Prevents the sender from overwhelming the receiver. Uses **receiver window (rwnd)** size advertised in ACKs.
6. **Congestion Control (TCP):** Prevents the sender from overwhelming the *network*. Uses algorithms like Slow Start, AIMD, Fast Retransmit. Also: Leaky Bucket and Token Bucket (traffic shaping).
7. **Multiplexing and Demultiplexing:**
 - **Multiplexing (sender):** Multiple application streams combined into one transport stream using port numbers.
 - **Demultiplexing (receiver):** Transport layer delivers data to the correct application process based on port number.

2 Port Addressing

Key Concept: Port Numbers

A **port number** is a 16-bit integer (0–65535) used to identify a specific process/application on a host. Combined with an IP address, it forms a **socket**: (IP address : port number).

Port number ranges:

Table 2.1: Port Number Classifications

Range	Name	Description
0 – 1023	Well-Known Ports	Assigned by IANA to standard services
1024 – 49151	Registered Ports	Used by vendor applications (MySQL: 3306, etc.)
49152 – 65535	Dynamic/Ephemeral	Temporarily assigned to client-side connections

Well-known port examples:

Table 2.2: Common Well-Known Port Numbers

Port	Protocol	Service
20/21	TCP	FTP (data/control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP (server/client)
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
443	TCP	HTTPS
3306	TCP	MySQL

3 UDP – User Datagram Protocol

Key Concept: UDP

UDP (RFC 768) is a **connectionless, unreliable** transport protocol. It provides a minimal service: process-to-process delivery with optional error checking. UDP adds very little overhead to IP – just an 8-byte header.

UDP Header Format:

[Source Port (16) | Dest Port (16) | Length (16) | Checksum (16)]
} 8 bytes

- **Source Port:** Sender's port (optional; 0 if not used).
- **Destination Port:** Receiver's port.
- **Length:** Total length (header + data) in bytes; minimum 8.
- **Checksum:** Optional error detection over header + data + pseudo-header.

UDP characteristics:

- No connection setup – send immediately.
- No guarantee of delivery, ordering, or duplicate protection.
- No flow control or congestion control.
- Very low overhead (8-byte header vs. TCP's 20-byte minimum).
- **Message-oriented:** Application sends a message; UDP delivers it as one datagram (no merging/splitting).

Table 3.1: UDP – Advantages and Disadvantages

Advantages	Disadvantages
Very fast (no handshake delay)	No guarantee of delivery
Low overhead (8-byte header)	No ordering guarantee
Supports broadcast and multicast	No flow/congestion control
Suitable for real-time apps	Application must handle reliability if needed

Applications using UDP: DNS, DHCP, SNMP, VoIP, video streaming (RTP), online gaming, TFTP, NTP.

4 TCP – Transmission Control Protocol

Key Concept: TCP

TCP (RFC 793) is a **connection-oriented, reliable, byte-stream** transport protocol. It guarantees that all data arrives at the destination *correctly, in order, and without duplicates*.

4.1 TCP Header Format

Table 4.1: TCP Header Fields

Field	Size	Description
Source Port	16 bits	Sender's port number
Destination Port	16 bits	Receiver's port number
Sequence Number	32 bits	Byte offset of first byte in this segment
Acknowledgement Number	32 bits	Next byte expected from the other side
Data Offset (HLEN)	4 bits	TCP header length in 32-bit words (min 5 = 20 bytes)
Reserved	6 bits	Must be zero
Control Flags	6 bits	URG, ACK, PSH, RST, SYN, FIN
Window Size	16 bits	Receiver buffer space available (flow control)
Checksum	16 bits	Error detection over header + data + pseudo-header
Urgent Pointer	16 bits	Offset to urgent data (if URG flag set)
Options	Variable	MSS, window scale, SACK, timestamps

TCP Control Flags:

- **SYN:** Synchronize sequence numbers (connection setup).
- **ACK:** Acknowledgement field is valid.
- **FIN:** Sender has finished sending data (connection teardown).
- **RST:** Reset the connection (abort).
- **PSH:** Push data to application immediately (no buffering).
- **URG:** Urgent data present; urgent pointer is valid.

4.2 TCP 3-Way Handshake (Connection Establishment)

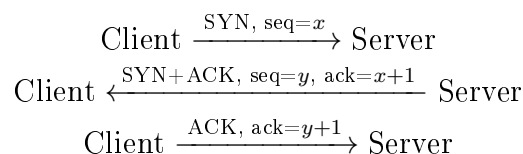
Key Concept: 3-Way Handshake

Before any data is exchanged, TCP establishes a connection using a **3-way handshake** to synchronize sequence numbers and confirm both parties are ready to communicate.

Steps:

1. **SYN** (Client → Server): Client sends segment with SYN flag set and its Initial Sequence Number (ISN): $\text{seq} = x$.
2. **SYN-ACK** (Server → Client): Server acknowledges with $\text{ack} = x + 1$ (next expected byte) and sends its own ISN: $\text{seq} = y$. SYN and ACK flags both set.

3. **ACK** (Client $\rightarrow$ Server): Client acknowledges server's SYN with $\text{ack} = y + 1$. ACK flag set. Connection is now established.



Why 3-way (not 2-way)? Two messages only confirm one direction. Three messages confirm that *both* sender-to-receiver and receiver-to-sender paths are working, and both sides agree on sequence numbers.

SYN flood attack: An attacker sends many SYN packets without completing the handshake, exhausting the server's connection table (half-open connections). Defence: SYN cookies.

4.3 TCP 4-Way Handshake (Connection Termination)

Key Concept: 4-Way Handshake

TCP connections are **full-duplex**; each direction must be closed independently. This requires a **4-way FIN handshake** (or sometimes a simultaneous close with 3 messages).

Steps:

1. **FIN** (Client $\rightarrow$ Server): Client has no more data to send. Sends FIN, $\text{seq} = u$.
2. **ACK** (Server $\rightarrow$ Client): Server acknowledges: $\text{ack} = u + 1$. Server may still send data (half-close).
3. **FIN** (Server $\rightarrow$ Client): Server finishes sending data; sends FIN, $\text{seq} = w$.
4. **ACK** (Client $\rightarrow$ Server): Client acknowledges: $\text{ack} = w + 1$. Client enters **TIME_WAIT** state for $2 \times \text{MSL}$ (Maximum Segment Lifetime $\approx 60\text{--}120$ s) before fully closing.

TIME_WAIT: Ensures the final ACK arrives at the server. If the server retransmits FIN (because ACK was lost), the client can re-send ACK. Also ensures old duplicate segments from this connection cannot affect a new connection on the same port.

4.4 TCP Reliability Mechanisms

- **Sequence numbers:** Every byte of data has a unique sequence number enabling reordering and duplicate detection.
- **Cumulative ACKs:** ACK n means all bytes up to $n - 1$ received correctly.
- **Retransmission timeout (RTO):** Dynamically estimated from RTT:

$$\text{RTT}_{\text{est}} = (1 - \alpha)\text{RTT}_{\text{est}} + \alpha \cdot \text{RTT}_{\text{sample}} \quad (\alpha \approx 0.125)$$

$$\text{RTO} = \text{RTT}_{\text{est}} + 4 \cdot \text{DevRTT}$$

- **Fast Retransmit:** If three duplicate ACKs received, retransmit the segment immediately (before RTO expires).

- **Byte-stream:** TCP provides an ordered stream of bytes; application receives data in the same order it was sent.

4.5 TCP Flow Control

Key Concept: TCP Flow Control

TCP uses a **sliding window** mechanism to prevent the sender from overflowing the receiver's buffer. The receiver advertises its available buffer space as the **receive window (rwnd)** in each ACK.

Sender may have at most $\min(\text{rwnd}, \text{cwnd})$ bytes outstanding

where **cwnd** = congestion window (see next section).

If **rwnd** = 0, sender pauses and periodically sends 1-byte *window probe* segments.

5 Congestion Control

Definition: Congestion

Congestion occurs when the offered traffic exceeds the capacity of the network (routers' buffers overflow), leading to packet loss and increased delay. Congestion control regulates the *sending rate* to prevent or alleviate congestion.

5.1 TCP Congestion Control

TCP uses an end-to-end congestion control algorithm based on the **congestion window (cwnd)**.

5.1.1 Slow Start

- Initially, **cwnd** = 1 MSS (Maximum Segment Size).
- For each ACK received, **cwnd increases by 1 MSS**.
- This doubles cwnd every RTT (exponential growth).
- Growth continues until cwnd reaches **ssthresh** (slow start threshold).

5.1.2 Congestion Avoidance (AIMD)

- When $\text{cwnd} \geq \text{ssthresh}$, switch to **additive increase**: cwnd increases by $1 \text{ MSS}^2 / \text{cwnd}$ per ACK (effectively +1 MSS per RTT).
- On **3 duplicate ACKs** (mild congestion): $\text{ssthresh} = \text{cwnd}/2$; $\text{cwnd} = \text{ssthresh}$ (Fast Recovery, TCP Reno).
- On **timeout** (severe congestion): $\text{ssthresh} = \text{cwnd}/2$; $\text{cwnd} = 1 \text{ MSS}$ (back to Slow Start).

5.2 Leaky Bucket Algorithm

Key Concept: Leaky Bucket

The **Leaky Bucket** algorithm shapes bursty input traffic into a *smooth, constant-rate* output stream. It models a bucket with a hole: water (packets) can enter at any rate but drains at a *constant rate*.

Algorithm:

1. Incoming packets are added to a bucket (FIFO queue) of capacity B bytes.
2. The bucket drains (outputs) at a constant rate of R bytes/second.
3. If the bucket is full, incoming packets are **discarded** (overflow).
4. Output is always at rate R regardless of input burstiness.

Parameters:

- B = bucket size (burst capacity) in bytes.
- R = output rate (leak rate) in bytes/second.

Effect: Converts bursty input into a uniform output stream. Bursts up to B bytes are smoothed; larger bursts cause packet loss.

Limitation: Always outputs at rate R even when the network is idle – cannot utilize spare capacity. This is addressed by Token Bucket.

Formula: Leaky Bucket Transmission Time

For a burst of n packets each of size L bytes, with bucket size B and output rate R :

$$T = \frac{B}{R} + \frac{(n - B/L) \cdot L}{R} = \frac{n \cdot L}{R}$$

The peak rate equals the output rate R (no burstiness in output).

5.3 Token Bucket Algorithm

Key Concept: Token Bucket

The **Token Bucket** algorithm allows *controlled bursts* while enforcing a long-term average rate. Tokens are added to a bucket at a constant rate r ; each packet (or byte) transmitted consumes one (or L) tokens. Transmission is allowed only if sufficient tokens are available.

Algorithm:

1. Tokens are generated at rate r tokens/second and added to a bucket.
2. Bucket capacity = b tokens (maximum burst size).
3. If the bucket is full, new tokens are **discarded** (not packets).
4. To transmit a packet, the sender must remove tokens from the bucket (1 token per byte, or 1 token per packet depending on implementation).
5. If insufficient tokens: packet must **wait** (or be discarded).

Formula: Token Bucket Maximum Burst

If tokens accumulate for time t seconds (starting from a full bucket):

$$\text{Max bytes sent} = b + r \cdot t$$

For a link of capacity M bytes/second, the maximum burst duration is:

$$t_{\max} = \frac{b}{M - r}$$

Worked Example: Token Bucket Example

Bucket capacity $b = 1000$ bytes, token rate $r = 500$ bytes/s, line rate $M = 2000$ bytes/s.

Max burst at start (full bucket): 1000 bytes at 2000 bytes/s = 0.5 s.

After burst, sustainable rate = $r = 500$ bytes/s.

Max burst duration: $t = 1000/(2000 - 500) = 0.67$ s.

Table 5.1: Leaky Bucket vs. Token Bucket

Property	Leaky Bucket	Token Bucket
Output rate	Constant (R) always	Variable (up to line rate)
Burst handling	Absorbed up to B ; excess dropped	Absorbed up to b tokens; allows bursts
Idle time	Still drains at R	Tokens accumulate (unused capacity saved)
Overflow	Packets discarded (full bucket)	Tokens discarded (packets buffered/dropped)
Use case	CBR traffic shaping (ATM)	VBR/Internet traffic policing
TCP compliance	Strict rate limiting	More flexible for TCP bursts

6 TCP vs. UDP – Complete Comparison

Table 6.1: TCP vs. UDP – Comprehensive Comparison

Property	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed delivery (ACK + retransmit)	No guarantee
Ordering	In-order delivery (sequence numbers)	Not guaranteed
Error control	Yes (checksum + retransmit)	Checksum only (optional)
Flow control	Yes (rwnd sliding window)	No
Congestion control	Yes (Slow Start, AIMD)	No
Header size	20–60 bytes	8 bytes
Speed	Slower (overhead)	Faster (minimal overhead)
Transmission type	Byte stream	Datagram (message boundaries)
Broadcast/Multicast	No	Yes
Applications	HTTP, FTP, SMTP, SSH	DNS, VoIP, video streaming, DHCP
RFC	793	768

7 Practice Questions (MAKAUT Style)

Short Answer (2–3 marks)

- Q1. What are the functions of the Transport Layer?
- Q2. What is a socket? How is it different from a port?
- Q3. What is the purpose of a 3-way handshake in TCP?
- Q4. Why is a 4-way handshake needed to terminate a TCP connection?
- Q5. What is the difference between flow control and congestion control?
- Q6. Compare TCP and UDP with at least 5 differences.
- Q7. What is the leaky bucket algorithm? State its limitations.
- Q8. What is the token bucket algorithm? How does it differ from leaky bucket?
- Q9. What are well-known ports? Give five examples.
- Q10. What is the purpose of the TIME_WAIT state in TCP?

Long Answer (5–6 marks)

- Q1.** Explain the TCP 3-way and 4-way handshaking processes with diagrams.
- Q2.** Explain the TCP header format. What is the role of each flag?
- Q3.** Explain the Leaky Bucket and Token Bucket algorithms. Compare them with a table.
- Q4.** Explain TCP's congestion control mechanism (Slow Start and AIMD). Draw the cwnd evolution graph.
- Q5.** A token bucket has a capacity of 2000 bytes and token generation rate of 1000 bytes/s. If a burst of 5000 bytes arrives, how long does it take to send if the output rate is 4000 bytes/s? Show all calculations.

8 Module 5 – Key Takeaways

Summary Checklist

- ✓ Transport layer: process-to-process delivery using port numbers (0–65535).
- ✓ Socket = IP address + port number; identifies a process uniquely.
- ✓ TCP: connection-oriented, reliable, ordered, flow+congestion control.
- ✓ UDP: connectionless, unreliable, fast, 8-byte header.
- ✓ 3-way handshake: SYN → SYN-ACK → ACK.
- ✓ 4-way teardown: FIN → ACK → FIN → ACK; TIME_WAIT = 2×MSL.
- ✓ TCP flow control: rwnd advertised by receiver.
- ✓ TCP congestion: Slow Start (exponential) → Congestion Avoidance (linear).
- ✓ Leaky Bucket: constant output rate R ; excess packets dropped.
- ✓ Token Bucket: allows bursts; max burst = $b + rt$; tokens accumulate when idle.

Module 6 — The Application Layer

DNS, HTTP/HTTPS, FTP, SMTP, POP3, IMAP, SSH, Telnet, SNMP

8 Hours · 10 Marks

1 Application Layer – Overview

Definition: Application Layer

The **Application Layer** (OSI Layer 7 / TCP/IP Layer 4) is the topmost layer, providing *network services directly to user applications*. It defines protocols for communication between applications and provides an interface to the underlying transport services (TCP/UDP).

Application layer paradigms:

- **Client-Server:** A server provides services; clients request them. Server has a fixed, well-known port. Examples: HTTP, FTP, DNS.
- **Peer-to-Peer (P2P):** Every node can act as both client and server. Examples: BitTorrent, Gnutella.

2 DNS – Domain Name System

Key Concept: DNS

DNS (RFC 1034/1035) is a *distributed, hierarchical naming system* that translates human-readable **domain names** (e.g., `www.google.com`) into **IP addresses** (e.g., `142.250.190.4`) – and vice versa (reverse DNS). It uses **UDP port 53** (and TCP for large responses/zone transfers).

2.1 DNS Hierarchy

Domain Name Space: Inverted tree structure.

Root (‘.’) → TLD → Second-Level Domain → Subdomain → Host

Example: `mail.cs.university.edu`.

`edu` = TLD, `university` = 2nd-level, `cs` = subdomain, `mail` = host.

DNS Server types:

- **Root Name Servers:** 13 sets (A through M) worldwide; know addresses of all TLD servers. Top of the hierarchy.
- **TLD (Top-Level Domain) Servers:** Manage TLDs like `.com`, `.edu`, `.org`, `.in`, etc.

- **Authoritative Name Servers:** Hold the actual DNS records for a specific domain (e.g., `google.com`).
- **Local (Recursive) Resolver:** DNS server used by a client (typically provided by ISP or configured as `8.8.8.8`). Performs the full recursive query on behalf of the client.

2.2 DNS Resolution Process

Recursive query (client to resolver):

1. Client asks local resolver: “What is the IP of `www.example.com`?”
2. Resolver checks its **cache**. If found, returns immediately.
3. If not cached, resolver queries a root server: “Who knows `.com`?”
4. Root returns address of `.com` TLD server.
5. Resolver queries `.com` TLD: “Who knows `example.com`?”
6. TLD returns address of authoritative server for `example.com`.
7. Resolver queries authoritative server: “What is `www.example.com`?”
8. Authoritative server returns the IP address.
9. Resolver caches the result (TTL) and returns IP to client.

DNS Record Types:

Table 2.1: Common DNS Resource Record (RR) Types

Type	Description
A	Maps hostname to IPv4 address
AAAA	Maps hostname to IPv6 address
CNAME	Canonical name / alias (points to another hostname)
MX	Mail exchange server for a domain
NS	Name server for a domain
PTR	Reverse DNS: IP address to hostname
SOA	Start of Authority: primary NS info for zone
TXT	Text records (used for SPF, DKIM, verification)

DNS Caching: All DNS responses include a **TTL (Time-to-Live)** value. Resolvers cache responses for TTL seconds to reduce DNS query load.

3 HTTP and HTTPS

3.1 HTTP – HyperText Transfer Protocol

Key Concept: HTTP

HTTP (RFC 7230–7235) is the foundation of data communication on the World Wide Web. It is a **stateless, application-layer protocol** operating over **TCP port 80**. The client (browser) sends *requests*; the server sends *responses*.

HTTP Request Methods:

Table 3.1: HTTP Methods

Method	Description
GET	Retrieve a resource (URL parameters in header)
POST	Submit data to the server (body contains data)
PUT	Replace a resource entirely
PATCH	Partial update of a resource
DELETE	Remove a resource
HEAD	Same as GET but only returns headers (no body)
OPTIONS	Query supported methods for a resource

HTTP Response Status Codes:

Table 3.2: HTTP Status Code Categories

Code	Category	Examples
1xx	Informational	100 Continue
2xx	Success	200 OK, 201 Created
3xx	Redirection	301 Moved Permanently, 302 Found
4xx	Client Error	400 Bad Request, 403 Forbidden, 404 Not Found
5xx	Server Error	500 Internal Server Error, 503 Service Unavailable

HTTP versions:

- **HTTP/1.0:** New TCP connection for each request/response.
- **HTTP/1.1:** *Persistent connections* (keep-alive); pipelining.
- **HTTP/2:** Binary framing; multiplexing (multiple requests on one connection); header compression; server push.
- **HTTP/3:** Uses QUIC (UDP-based) instead of TCP; lower latency.

HTTP is stateless: Each request is independent. State management uses **cookies** (client-side) and **sessions** (server-side).

3.2 HTTPS – HTTP Secure

Key Concept: HTTPS

HTTPS is **HTTP over TLS/SSL** (Transport Layer Security). It encrypts all communication between client and server, providing **confidentiality**, **integrity**, and **authentication**. Operates on **TCP port 443**.

TLS Handshake (simplified):

1. Client sends **ClientHello** (supported TLS versions, cipher suites, random).
2. Server sends **ServerHello** + **Certificate** (X.509, with public key).
3. Client verifies certificate against a trusted CA.
4. Client generates pre-master secret, encrypts with server's public key.
5. Both derive session keys from pre-master secret.
6. Encrypted communication begins.

Difference HTTP vs HTTPS:

Property	HTTP	HTTPS
Port	80	443
Encryption	None	TLS/SSL
Authentication	None	Server certificate (CA-signed)
Data integrity	None	MAC/HMAC in TLS
Use case	Non-sensitive web	E-commerce, banking, login

4 FTP – File Transfer Protocol

Key Concept: FTP

FTP (RFC 959) transfers files between a client and server over **TCP**. It uses **two separate connections**:

- **Control connection** (port 21): Persists for the session; carries commands (USER, PASS, LIST, RETR, STOR, QUIT) and responses.
- **Data connection** (port 20 in active mode): Opened for each file transfer; closed when transfer is complete.

Active vs. Passive FTP:

Table 4.1: Active vs. Passive FTP

Property	Active FTP	Passive FTP
Data connection	Server connects back to client (port 20)	Client connects to server (random high port)
Firewall issues	Problems if client is behind firewall	Preferred; client initiates both connections
Client port	Client sends PORT command with its IP:port	Client sends PASV; server provides its port

FTP commands: USER, PASS, LIST, RETR (download), STOR (upload), MKD, RMD, QUIT.
Limitation: Credentials and data transmitted in **plaintext**. Use **FTPS** (FTP over TLS) or **SFTP** (SSH File Transfer Protocol) for security.

5 Email Protocols: SMTP, POP3, IMAP

5.1 SMTP – Simple Mail Transfer Protocol

Key Concept: SMTP

SMTP (RFC 5321) is the standard protocol for **sending and relaying** email. It operates over **TCP port 25** (server-to-server) or **port 587** (client submission). SMTP is a *push* protocol.

SMTP communication flow:

1. **Sender's mail client** sends email to **sender's SMTP server** using SMTP (port 587).
2. Sender's SMTP server looks up **MX record** of recipient domain via DNS.
3. Sender's server relays email to **recipient's SMTP server** (port 25).
4. Recipient retrieves email from their mail server using **POP3 or IMAP**.

SMTP Commands: HELO/EHLO, MAIL FROM, RCPT TO, DATA, QUIT, AUTH.

SMTP Extensions (ESMTP): STARTTLS for encryption, AUTH for authentication.

5.2 POP3 – Post Office Protocol v3

Key Concept: POP3

POP3 (RFC 1939) is used by email clients to **download** messages from a mail server. Operates over **TCP port 110** (or 995 with TLS). Messages are typically **deleted from the server** after download (though “leave on server” option exists).

POP3 Phases:

1. **Authorization:** USER + PASS to authenticate.

2. **Transaction:** LIST (list messages), RETR *n* (retrieve message *n*), DELE *n* (mark for deletion), STAT.
3. **Update:** QUIT – deletions applied, connection closed.

Limitation: Designed for single-device access. After download, emails not accessible from other devices.

5.3 IMAP – Internet Message Access Protocol

Key Concept: IMAP

IMAP (RFC 3501) provides **online access** to email stored on the server. Operates over **TCP port 143** (or 993 with TLS). Emails remain on the server; client *synchronizes* with the server – enabling multi-device access.

Table 5.1: POP3 vs. IMAP

Property	POP3	IMAP
Port	110 (995 with TLS)	143 (993 with TLS)
Storage	Downloads to client; deletes from server	Stays on server; synced
Multi-device	Poor (emails on one device only)	Excellent (all devices in sync)
Offline access	Yes (after download)	Limited (needs server connection)
Folders	No server-side folders	Server-side folder management
Bandwidth	Download entire message	Downloads headers first; body on demand

6 Remote Login: Telnet and SSH

6.1 Telnet

Key Concept: Telnet

Telnet (RFC 854) provides a **remote terminal (virtual terminal)** service. It allows a user to log into a remote host and execute commands as if sitting at that machine. Uses **TCP port 23**.

Features:

- Uses **NVT (Network Virtual Terminal)** to standardize terminal representation.
- Client and server negotiate terminal options.
- **Critical security flaw:** All data (including username and password) transmitted in **plaintext** – easily intercepted.

- Largely replaced by **SSH** in modern networks.
- Still used for testing/debugging (e.g., `telnet host 80` to test HTTP manually).

6.2 SSH – Secure Shell

Key Concept: SSH

SSH (RFC 4251) is a **cryptographic network protocol** that provides a *secure* channel over an unsecured network. It is the modern, secure replacement for Telnet and FTP. Uses **TCP port 22**.

SSH provides:

- **Secure remote terminal access** (replaces Telnet).
- **Secure file transfer** – SFTP and SCP (replaces FTP).
- **Port forwarding / tunneling** – forwards TCP connections securely.

SSH Authentication methods:

- **Password authentication:** Password sent over encrypted channel.
- **Public-key authentication:** Client has a key pair (private + public). Public key placed on server. Server challenges client; only client with matching private key can respond.

SSH Architecture (3 protocols in SSH-2):

1. **Transport Layer Protocol:** Key exchange (Diffie-Hellman), encryption (AES), integrity (HMAC-SHA).
2. **User Authentication Protocol:** Authenticates the user to the server.
3. **Connection Protocol:** Multiplexes the encrypted channel into multiple logical channels (terminal, file transfer, forwarding).

Table 6.1: Telnet vs. SSH

Property	Telnet	SSH
Port	23	22
Security	None (plaintext)	Encrypted (AES, ChaCha20, etc.)
Authentication	Username/password (plaintext)	Password or public-key
Integrity	None	HMAC
Use today	Rarely (legacy/debugging)	Standard for remote access

7 File Sharing: NFS and SMB

7.1 NFS – Network File System

Key Concept: NFS

NFS (Sun Microsystems, RFC 7530) allows a client to access files on a remote server as if they were on a local file system. It uses the **client-server model** and is dominant in **Unix/Linux** environments. Uses **UDP or TCP port 2049**.

NFS Architecture:

- **Server:** Exports (shares) parts of its file system.
- **Client:** Mounts the exported file system at a local mount point.
- **RPC (Remote Procedure Call):** NFS operations (READ, WRITE, GETATTR, LOOKUP) are implemented as RPC calls.
- **Stateless (NFSv2/3):** Server does not maintain session state; each request is self-contained. NFSv4 is stateful.

NFS versions:

- **NFSv2/v3:** UDP-based, stateless.
- **NFSv4:** TCP-based, stateful, better security (Kerberos), single port 2049, file locking built-in.

7.2 SMB – Server Message Block

Key Concept: SMB

SMB (also called **CIFS** – Common Internet File System) is the standard file-sharing protocol for **Windows** environments. It allows clients to access files, printers, and other resources on a network. Uses **TCP port 445** (direct) or **NetBIOS over TCP port 139**.

Features:

- Supports file read/write, directory browsing, file locking, and printing.
- **Authentication:** Supports NTLM and Kerberos.
- **Samba:** Open-source implementation of SMB for Linux/Unix, enabling Windows clients to access Linux shares.
- **SMB versions:** SMBv1 (legacy, vulnerable to WannaCry ransomware), SMBv2 (improved performance), SMBv3 (encryption support).

Table 7.1: NFS vs. SMB Comparison

Property	NFS	SMB
Origin	Sun Microsystems	IBM/Microsoft
Primary platform	Unix/Linux	Windows
Port	UDP/TCP 2049	TCP 445
Protocol base	RPC	Session-based
Authentication	UNIX UID/GID, NFSv4 + Kerberos	NTLM, Kerberos
Cross-platform	Linux-native; Windows needs NFS client	Windows-native; Samba on Linux

8 SNMP – Simple Network Management Protocol

Key Concept: SNMP

SNMP (RFC 3411–3418) is the standard protocol for **monitoring and managing** network devices (routers, switches, servers, printers) from a central management station. Uses **UDP port 161** (agent) and **UDP port 162** (trap receiver).

8.1 SNMP Architecture

Three components:

1. **Network Management Station (NMS) / Manager:** Runs SNMP management software; sends queries and receives traps.
2. **SNMP Agent:** Software running on each managed device; responds to NMS queries and sends traps on events.
3. **MIB (Management Information Base):** A virtual database of managed objects on each device. Each object has a unique **OID (Object Identifier)** in a hierarchical tree format.

8.2 SNMP Operations

Table 8.1: SNMP Protocol Operations

Operation	Description
GetRequest	NMS requests the value of one or more MIB objects from an agent
GetNextRequest	NMS traverses the MIB tree (walks to next object)
GetBulkRequest	NMS retrieves multiple MIB objects efficiently (SNMPv2+)
SetRequest	NMS sets the value of a MIB object on an agent (configuration)
GetResponse	Agent's reply to Get/Set requests
Trap	Agent proactively notifies NMS of an event (link down, threshold exceeded)
InformRequest	Like Trap but requires acknowledgement (SNMPv2+)

8.3 SNMP Versions

Table 8.2: SNMP Version Comparison

Property	SNMPv1		SNMPv2c		SNMPv3
Security	Community	string	Community	string	Username/password + encryption
	(plaintext)		(plaintext)		
Authentication	None		None		HMAC-MD5 or HMAC-SHA
Encryption	None		None		DES or AES
GetBulk	No		Yes		Yes
Inform	No		Yes		Yes

Community strings (SNMPv1/v2c):

- **Read-Only community:** “public” (NMS can query but not modify).
- **Read-Write community:** “private” (NMS can query and modify).
- Transmitted in **plaintext** – a significant security weakness. Use SNMPv3 in production.

9 Application Protocol Summary Table

Table 9.1: Complete Application Layer Protocol Reference

Protocol	Port	Transport	Key Function
DNS	53	UDP/TCP	Domain name to IP resolution
HTTP	80	TCP	Web page transfer
HTTPS	443	TCP	Encrypted web (HTTP+TLS)
FTP	20/21	TCP	File transfer (data/control)
SMTP	25/587	TCP	Email sending/relay
POP3	110/995	TCP	Email download (delete from server)
IMAP	143/993	TCP	Email sync (stays on server)
Telnet	23	TCP	Unencrypted remote terminal
SSH	22	TCP	Encrypted remote terminal + file transfer
SNMP	161/162	UDP	Network device management
DHCP	67/68	UDP	Dynamic IP assignment
NFS	2049	UDP/TCP	Unix/Linux network file system
SMB/CIFS	445	TCP	Windows network file/printer sharing

10 Practice Questions (MAKAUT Style)

Short Answer (2–3 marks)

- Q1.** What is DNS? Explain the DNS resolution process.
- Q2.** What are DNS record types? Explain A, CNAME, MX, and NS records.
- Q3.** What is HTTP? How does it differ from HTTPS?
- Q4.** What are HTTP status codes? Give two examples from each category.
- Q5.** Explain the two connections used in FTP. What is the difference between active and passive FTP?
- Q6.** What is the difference between POP3 and IMAP? Which is preferred for multi-device use?
- Q7.** What is SMTP? Explain the email delivery process from sender to recipient.
- Q8.** Compare Telnet and SSH. Why is SSH preferred?
- Q9.** What is SNMP? Name its three main components.
- Q10.** What is NFS? How does it differ from SMB?

Long Answer (5–6 marks)

- Q1.** Explain the DNS hierarchy and the step-by-step process of resolving `www.makaut.ac.in`. Include root servers, TLD servers, and authoritative servers.
- Q2.** Explain HTTP with its request methods, response status codes, and the differences between HTTP/1.0, HTTP/1.1, and HTTP/2.
- Q3.** Describe the FTP protocol in detail: connections, commands, active vs passive modes, and security considerations.
- Q4.** Explain SMTP, POP3, and IMAP. How do they work together in an email system? Compare POP3 and IMAP.
- Q5.** Explain SNMP architecture (Manager, Agent, MIB). Describe all SNMP operations and compare SNMPv1, v2c, and v3.
- Q6.** Compare NFS and SMB. What is Samba? Explain the role of each in a mixed OS environment.

11 Module 6 – Key Takeaways

Summary Checklist

- ✓ DNS: UDP/53; hierarchy: root → TLD → authoritative; caches with TTL.
- ✓ DNS records: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), NS (name-server).
- ✓ HTTP: TCP/80; stateless; methods GET/POST/PUT/DELETE; versions 1.0/1.1/2/3.
- ✓ HTTPS: TCP/443; HTTP + TLS; provides confidentiality, integrity, authentication.
- ✓ FTP: TCP/21 (control), TCP/20 (data); active vs passive mode.
- ✓ SMTP: TCP/25 or 587; sends mail. POP3: TCP/110, downloads. IMAP: TCP/143, syncs.
- ✓ Telnet: TCP/23; plaintext; insecure. SSH: TCP/22; encrypted; replaces Telnet+FTP.
- ✓ NFS: Unix/Linux, UDP/TCP 2049, RPC-based. SMB: Windows, TCP/445; Samba for Linux.
- ✓ SNMP: UDP/161 (agent), 162 (trap); Manager–Agent–MIB architecture.
- ✓ SNMP operations: GetRequest, SetRequest, Trap (agent-initiated), InformRequest.
- ✓ SNMPv3 adds security (authentication + encryption); v1/v2 use community strings.

Module 7 — Emerging Concepts

Cloud Networking, IoT, MQTT, CoAP, Software-Defined Networking (SDN)

6 Hours · 6 Marks

1 Cloud Networking Basics

1.1 What is Cloud Computing?

Definition: Cloud Computing

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. Users pay only for what they use (*pay-as-you-go* model).

The term “cloud” refers to the Internet, traditionally depicted as a cloud in network diagrams. Cloud computing abstracts physical infrastructure and presents it as an on-demand service.

Key characteristics of cloud computing (NIST definition):

1. **On-demand self-service:** Users can provision resources (compute, storage) automatically without human interaction with the provider.
2. **Broad network access:** Resources accessible over the network from any device (laptop, phone, tablet).
3. **Resource pooling:** Provider resources are pooled to serve multiple consumers (multi-tenancy) using virtualization.
4. **Rapid elasticity:** Resources can be scaled up or down quickly (auto-scaling) to match demand.
5. **Measured service:** Resource usage is monitored, controlled, and reported transparently (metered billing).

1.2 Cloud Deployment Models

Table 1.1: Cloud Deployment Models Comparison

Model	Definition	Advantages	Disadvantages
Public Cloud	Infrastructure owned and operated by a third-party provider (AWS, Azure, GCP); shared among multiple tenants	Low cost, no maintenance, highly scalable	Less control, potential security concerns
Private Cloud	Infrastructure dedicated to a single organization; on-premises or hosted	Full control, better security, compliance	High cost, limited scalability
Hybrid Cloud	Combination of public and private clouds; workloads can move between them	Flexible, balance cost and security	Complex to manage, integration challenges
Community Cloud	Shared by organizations with common concerns (e.g., government, healthcare)	Shared cost, meets specific compliance	Limited to community members

1.3 Cloud Service Models

Key Concept: IaaS – PaaS – SaaS

Cloud services are delivered in three fundamental models, often described as a *pizza analogy*: from “made from scratch” (on-premises) to fully delivered (SaaS).

1.3.1 IaaS – Infrastructure as a Service

Definition: IaaS

IaaS provides virtualized computing infrastructure (virtual machines, storage, networking) over the Internet. The user manages the OS, middleware, and applications; the provider manages the physical hardware and hypervisor.

- User controls: OS, runtime, middleware, applications, data.

- Provider controls: Servers, storage, networking, hypervisor.
- **Examples:** Amazon EC2, Microsoft Azure VMs, Google Compute Engine.
- **Use cases:** Test and development environments, big data analysis, high-performance computing.

1.3.2 PaaS – Platform as a Service

Definition: PaaS

PaaS provides a platform (runtime environment, databases, middleware, development tools) allowing developers to build, deploy, and manage applications without managing the underlying infrastructure.

- User controls: Applications and data only.
- Provider controls: Infrastructure + OS + runtime + middleware.
- **Examples:** Google App Engine, Microsoft Azure App Service, Heroku, AWS Elastic Beanstalk.
- **Use cases:** Application development, API development, microservices.

1.3.3 SaaS – Software as a Service

Definition: SaaS

SaaS delivers complete software applications over the Internet on a subscription basis. Users access the software via a web browser; the provider manages everything (infrastructure, platform, and application).

- User controls: Only usage and data input.
- Provider controls: Everything else.
- **Examples:** Gmail, Microsoft 365, Salesforce, Zoom, Dropbox.
- **Use cases:** Email, collaboration, CRM, ERP, communication.

Table 1.2: IaaS vs. PaaS vs. SaaS – Responsibility Stack

Component	IaaS	PaaS	SaaS
Applications	User	User	Provider
Data	User	User	Provider
Runtime	User	Provider	Provider
Middleware	User	Provider	Provider
OS	User	Provider	Provider
Virtualization	Provider	Provider	Provider
Servers	Provider	Provider	Provider
Storage	Provider	Provider	Provider
Networking	Provider	Provider	Provider

Blue = User-managed Red = Provider-managed

1.4 Cloud Networking Architecture

Key Concept: Cloud Networking

Cloud networking refers to the use of virtualized network resources — switches, routers, firewalls, load balancers — to connect cloud resources internally (within data centers) and externally (to users and on-premises systems).

Key cloud networking components:

- **VPC (Virtual Private Cloud):** Isolated virtual network within a public cloud. Users define IP ranges, subnets, routing tables, and security groups. Example: AWS VPC, Azure Virtual Network.
- **Virtual Switch (vSwitch):** Software-based switch connecting VMs within a hypervisor (e.g., Open vSwitch – OVS).
- **Overlay Networks:** Logical networks built on top of physical networks using tunneling protocols (VXLAN, GRE, NVGRE). Allow VM mobility across physical hosts without changing IP addresses.
- **Load Balancer:** Distributes incoming traffic across multiple server instances to ensure availability and performance.
- **CDN (Content Delivery Network):** Geographically distributed servers that cache content close to users to reduce latency.
- **VPN (Virtual Private Network):** Secure encrypted tunnel connecting remote users or on-premises data centers to the cloud.
- **Direct Connect / ExpressRoute:** Dedicated physical network connection bypassing the public Internet for higher reliability and lower latency.

1.5 Virtualization in Cloud Networking

Definition: Virtualization

Virtualization is the abstraction of physical hardware into multiple logical instances. A **hypervisor** (Virtual Machine Monitor, VMM) runs on physical hardware and hosts multiple **Virtual Machines (VMs)**, each with its own OS and network stack.

Types of hypervisors:

- **Type 1 (Bare-metal):** Runs directly on hardware. Examples: VMware ESXi, Microsoft Hyper-V, KVM, Xen.
- **Type 2 (Hosted):** Runs on top of a host OS. Examples: VMware Workstation, VirtualBox.

Network Function Virtualization (NFV): Replaces dedicated network hardware (firewalls, load balancers, IDS) with software running on standard servers. Reduces cost and increases flexibility.

Key cloud networking advantages:

- Elastic bandwidth allocation on demand.
- Global low-latency connectivity through provider backbone.
- Automated network provisioning (Infrastructure as Code).
- Pay-per-use billing.

2 Internet of Things (IoT) Networking

2.1 Definition and Overview

Definition: Internet of Things (IoT)

The **Internet of Things (IoT)** refers to the network of *physical objects* (“things”) — sensors, actuators, appliances, vehicles, industrial machines — embedded with electronics, software, sensors, and network connectivity, enabling them to **collect and exchange data** over the Internet without human intervention.

Note

Kevin Ashton coined the term “Internet of Things” in 1999 in the context of supply chain management using RFID tags. Today, IoT encompasses billions of connected devices across all sectors.

2.2 IoT Architecture

The IoT architecture is typically described as a **4-layer model**:

Layer 1: Perception Layer (Sensing Layer):

- Physical devices: sensors (temperature, humidity, pressure, GPS, camera), actuators (motors, valves, LEDs).
- Collects raw data from the physical world.
- Technologies: RFID, NFC, Bluetooth, Zigbee, Z-Wave.

Layer 2: Network Layer (Connectivity Layer):

- Transmits data from perception layer to the processing layer.
- Technologies: Wi-Fi, LTE/5G, LoRaWAN, NB-IoT, Zigbee, BLE, Ethernet, 6LoWPAN.
- Handles routing, addressing, and security during transmission.

Layer 3: Processing Layer (Middleware / Fog Layer):

- Stores, analyzes, and processes data from the network layer.
- Uses cloud computing, edge computing, or fog computing.
- Technologies: Cloud platforms (AWS IoT, Azure IoT Hub), edge gateways, databases, stream processing.

Layer 4: Application Layer:

- Delivers application-specific services to the end user.
- Examples: Smart home app, industrial dashboard, healthcare portal.
- Protocols: MQTT, CoAP, HTTP/REST, AMQP, WebSocket.

2.3 Key Characteristics of IoT

- **Heterogeneity:** Diverse devices with different hardware, OS, protocols, and capabilities.
- **Massive Scale:** Billions of connected devices; requires scalable networking protocols.
- **Resource Constraints:** Many IoT devices have limited CPU, memory, battery, and bandwidth – necessitating lightweight protocols.
- **Connectivity:** Always-on or periodic connectivity using wired or wireless technologies.
- **Real-time / Near-real-time:** Many IoT applications require low latency (e.g., industrial control, health monitoring).
- **Security and Privacy:** Large attack surface; devices often lack robust security mechanisms.

2.4 IoT Networking Challenges

- **Scalability:** Managing millions/billions of devices and their data.
- **Interoperability:** Devices from different vendors using different protocols must communicate.
- **Security:** Devices often lack computational power for strong encryption; prone to attacks (DDoS via botnet, data breaches).
- **Power consumption:** Battery-powered devices need very low power protocols (LoRaWAN, NB-IoT, Zigbee).

- **Bandwidth and Latency:** Huge data volumes from sensor networks; critical applications need ultra-low latency.
- **IP Addressing:** IPv4 exhaustion – IoT accelerates IPv6 adoption.
- **Data Management:** Volume, velocity, and variety of IoT data (Big Data challenges).

2.5 IoT Applications

Table 2.1: IoT Application Domains

Domain	Examples
Smart Home	Smart thermostats, lighting, security cameras, door locks, voice assistants
Healthcare	Wearable monitors (heart rate, glucose), remote patient monitoring, smart pills
Industrial IoT (IIoT)	Predictive maintenance, factory automation, supply chain tracking, SCADA
Smart City	Traffic management, smart parking, waste management, environmental monitoring
Agriculture	Soil sensors, precision irrigation, drone-based crop monitoring, livestock tracking
Transportation	Vehicle tracking (GPS/RFID), autonomous vehicles, fleet management
Energy	Smart meters, smart grid, demand response, renewable energy monitoring
Retail	Inventory tracking (RFID), smart shelves, cashierless check-out

2.6 IoT Wireless Technologies

Table 2.2: IoT Wireless Technology Comparison

Technology	Range	Data Rate	Power	Use Case
Wi-Fi (802.11)	100 m	High	High	Smart home, IP cameras
Bluetooth LE	10–100 m	Low–Med	Very Low	Wearables, beacons
Zigbee (802.15.4)	10–100 m	Low	Very Low	Home automation, mesh
Z-Wave	30 m	Low	Very Low	Home automation
LoRaWAN	2–15 km	Very Low	Very Low	Smart city, agriculture
NB-IoT	10+ km	Very Low	Very Low	Smart meters, tracking
6LoWPAN	10–100 m	Low	Very Low	IPv6 over low-power mesh
5G NR	1 km	Very High	Med	Industrial IoT, V2X

3 MQTT – Message Queuing Telemetry Transport

Key Concept: MQTT

MQTT (ISO/IEC 20922) is a lightweight, **publish-subscribe** messaging protocol designed for *constrained devices and low-bandwidth, high-latency, or unreliable networks*. It was originally developed by IBM (Andy Stanford-Clark and Arlen Nipper) for monitoring oil pipelines via satellite. It is now the de facto standard for IoT messaging.

3.1 MQTT Architecture: Publish-Subscribe Model

Definition: Publish-Subscribe

In the **publish-subscribe pattern**, message senders (*publishers*) do not send messages directly to receivers (*subscribers*). Instead, all messages are sent to a central **broker**, which routes them to subscribers based on *topic* subscriptions. Publishers and subscribers are *decoupled*: they do not know each other.

Three roles in MQTT:

1. **Publisher (Client)**: Sends (publishes) messages to the broker on a specific *topic* (e.g., `home/livingroom/temperature`).
2. **Broker (Server)**: Receives all messages, filters them by topic, and distributes them to all clients subscribed to that topic. Examples: Mosquitto, HiveMQ, AWS IoT Core, Azure IoT Hub.
3. **Subscriber (Client)**: Registers interest in one or more topics with the broker; receives messages whenever a publisher posts to those topics.

Communication flow:

Sensor (Publisher) $\xrightarrow{\text{PUBLISH: home/temp} = 25^{\circ}\text{C}}$ Broker $\xrightarrow{\text{DISTRIBUTE}}$ Dashboard (Subscriber)

3.2 MQTT Topics

- Topics are **hierarchical strings** separated by “/”: e.g., `building/floor2/room5/humidity`.
- **Wildcard +** (single level): matches one level. e.g., `building+/room5/humidity` matches `building/floor1/room5/humidity`.
- **Wildcard #** (multi-level): matches remaining levels. e.g., `building/floor2/#` matches all sensors on floor 2.

3.3 MQTT Quality of Service (QoS) Levels

Table 3.1: MQTT QoS Levels

Level	Name	Guarantee	Overhead
0	At most once (Fire and forget)	Message delivered at most once; no ACK; possible loss	Lowest
1	At least once	Message delivered at least once; ACK required; possible duplicates	Medium
2	Exactly once	Message delivered exactly once; 4-part handshake (PUBLISH, PUBREC, PUBREL, PUBCOMP)	Highest

Note

QoS 0 is used for sensor data where occasional loss is acceptable (e.g., temperature readings every second).

QoS 1 is used for commands where duplicates can be tolerated.

QoS 2 is used for billing or control commands where exactly-once delivery is critical.

3.4 MQTT Packet Types (Control Packets)

Table 3.2: MQTT Control Packet Types

Packet	Direction	Description
CONNECT	Client → Broker	Initiate connection; carries ClientID, credentials, keep-alive, LWT
CONNACK	Broker → Client	Connection acknowledgement; return code (0=success)
PUBLISH	Both directions	Send a message on a topic; carries topic, payload, QoS, retain flag
PUBACK	Receiver → Sender	ACK for QoS 1 PUBLISH
PUBREC	Receiver → Sender	Step 1 of QoS 2: PUBLISH received
PUBREL	Sender → Receiver	Step 2 of QoS 2: PUBLISH release
PUBCOMP	Receiver → Sender	Step 3 of QoS 2: PUBLISH complete
SUBSCRIBE	Client → Broker	Subscribe to one or more topics with QoS
SUBACK	Broker → Client	Subscription acknowledgement; confirms granted QoS
UNSUBSCRIBE	Client → Broker	Unsubscribe from topics
UNSUBACK	Broker → Client	Unsubscribe acknowledgement
PINGREQ	Client → Broker	Keep-alive ping
PINGRESP	Broker → Client	Keep-alive pong
DISCONNECT	Client → Broker	Graceful disconnection

3.5 MQTT Special Features

3.5.1 Last Will and Testament (LWT)

When a client connects, it can specify a **LWT message**: a topic and payload that the broker will automatically publish if the client *disconnects unexpectedly* (without sending DISCONNECT). This allows other clients to detect device failures.

3.5.2 Retained Messages

A PUBLISH message with the **retain flag** = **1** causes the broker to store the *last message* on that topic. New subscribers immediately receive this retained message, even if no publisher is currently active.

3.5.3 Persistent Sessions

If a client connects with **CleanSession** = **false**, the broker stores subscriptions and queued messages (QoS 1/2) for the client while it is offline. When the client reconnects,

it receives missed messages.

3.6 MQTT Transport and Security

- **Transport:** MQTT runs over **TCP** (default port **1883**).
- **Secure MQTT (MQTTS):** MQTT over **TLS** (port **8883**) for encryption and authentication.
- **MQTT over WebSocket:** Port 9001/443 for browser-based clients.
- **Authentication:** Username/password in CONNECT packet (but transmitted in plaintext without TLS – always use TLS in production).
- **Authorization:** ACL (Access Control Lists) configured on the broker control which clients can publish/subscribe to which topics.

Formula: MQTT Fixed Header Structure

Every MQTT packet starts with a 2-byte minimum fixed header:



Remaining length is encoded as a variable-length integer (1–4 bytes), using 7 bits per byte with MSB as continuation flag. This allows packets up to 268,435,455 bytes ($\approx$ 256 MB).

4 CoAP – Constrained Application Protocol

Key Concept: CoAP

CoAP (RFC 7252) is a specialized **RESTful** application-layer protocol designed for use in *constrained environments* (microcontrollers, low-power lossy networks). It mirrors the HTTP request-response model but is optimized for **machine-to-machine (M2M)** communication over **UDP**, making it extremely lightweight and suitable for IoT.

4.1 CoAP Design Principles

- **RESTful:** Uses the same four methods as HTTP (GET, POST, PUT, DELETE) enabling Web-of-Things integration.
- **UDP-based:** Runs over **UDP port 5683** (UDP/5684 with DTLS). Low overhead; no TCP connection setup.
- **Reliability over UDP:** Achieves reliability through built-in Confirmable messages and Acknowledgements (without TCP's complexity).
- **Asynchronous:** Supports non-blocking request-response.
- **Binary header:** Fixed 4-byte header (vs. HTTP's verbose text headers).
- **Multicast support:** Can send requests to a group of devices.
- **Low overhead:** Typical CoAP message = 10–20 bytes (vs. HTTP = hundreds).

4.2 CoAP Message Types

Table 4.1: CoAP Message Types

Type (2 bits)	Name	Description
00	CON (Confirmable)	Reliable message; sender retransmits until ACK received
01	NON (Non-confirmable)	Unreliable; sent once; no ACK required (like UDP)
10	ACK (Acknowledgement)	Acknowledges a CON message; may carry a response (piggybacked)
11	RST (Reset)	Indicates a CON/NON message could not be processed

4.3 CoAP Request-Response Model

CoAP Methods (same semantics as HTTP):

- **GET:** Retrieve the current value of a resource (e.g., read sensor).
- **POST:** Process the representation enclosed in the request (create/update resource or trigger action).
- **PUT:** Update or create a resource with the given representation.
- **DELETE:** Delete a resource.

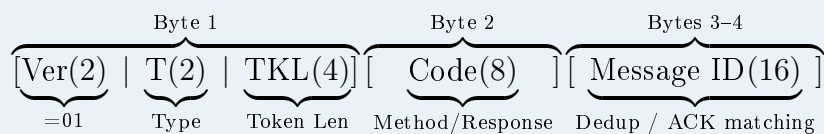
CoAP Response Codes (3-digit similar to HTTP):

Table 4.2: CoAP Response Codes

Code	Name	Meaning
2.05	Content	Successful GET response
2.01	Created	Resource created (POST/PUT)
2.04	Changed	Resource updated
4.04	Not Found	Resource not found
4.05	Method Not Allowed	Method not supported
5.00	Internal Server Error	Server error

4.4 CoAP Header Format

Formula: CoAP Fixed Header (4 bytes)



Followed by: Token (0–8 bytes), Options (variable), Payload (variable).

Message ID: 16-bit identifier used to match ACK to CON, and to detect duplicate NON messages.

Token: Used to match responses to requests across multiple exchanges.

4.5 CoAP Advanced Features

4.5.1 Observe Extension (RFC 7641)

Allows a client to **subscribe to a resource** and receive notifications when its value changes – similar to MQTT’s publish-subscribe but over CoAP. The client sends a GET with the **Observe** option; the server sends notifications (NON or CON) whenever the resource changes.

4.5.2 Block-wise Transfer (RFC 7959)

Since CoAP uses UDP, large payloads must be split into **blocks**. Block-wise transfer allows transferring large resources (firmware updates, large sensor data) in multiple CoAP transactions.

4.5.3 CoAP over DTLS (Secure CoAP)

CoAP uses **DTLS** (Datagram TLS – TLS adapted for UDP) for security, providing encryption and authentication. Uses port **5684**. The URI scheme is **coaps://** (analogous to HTTPS).

4.6 MQTT vs. CoAP – Detailed Comparison

Table 4.3: MQTT vs. CoAP Comprehensive Comparison

Property	MQTT	CoAP
Full Name	Message Queuing Telemetry Transport	Constrained Application Protocol
Architecture	Publish-Subscribe (via broker)	Request-Response (RESTful, client-server)
Transport	TCP (port 1883 / 8883 with TLS)	UDP (port 5683 / 5684 with DTLS)
Header size	2 bytes minimum	4 bytes fixed
Reliability	QoS 0/1/2 levels	CON/NON message types
Message pattern	Event-driven (push)	Pull or push (Observe extension)
Broker/Intermediary	Required (broker is central)	Not required (direct device-to-device)
Security	TLS (MQTTS, port 8883)	DTLS (CoAPS, port 5684)
Multicast	Not supported	Supported (UDP multicast)
Resource discovery	No built-in	Yes (CoRE Link Format, RFC 6690)
Web integration	Requires adaptation	Native REST (maps to HTTP easily)
Best for	Many-to-many, event-driven, unreliable networks	Device-to-device, web integration, constrained devices
Typical devices	Raspberry Pi, smartphones, gateways	Microcontrollers (Arduino, CC2538)
Standards body	OASIS, ISO/IEC 20922	IETF RFC 7252

5 Software-Defined Networking (SDN)

5.1 Motivation for SDN

Traditional networking has significant limitations:

- **Tightly coupled control and data planes:** Each network device (router, switch) has its own control software embedded in proprietary hardware – making management complex and inflexible.

- **Vendor lock-in:** Different vendors use proprietary protocols and management interfaces.
- **Static configuration:** Network policies are configured device-by-device – slow to adapt to changing traffic patterns.
- **Complexity:** Modern networks run dozens of distributed protocols (OSPF, RSTP, VLAN, QoS, ACLs) – difficult to troubleshoot.

Key Concept: Software-Defined Networking (SDN)

SDN is a network architecture that **decouples the control plane from the data plane**. The *control plane* (routing decisions, policies) is moved to a centralized **SDN Controller**, while the *data plane* (packet forwarding) remains in the network devices. The controller programs the forwarding rules in devices via a **southbound API** (e.g., OpenFlow).

5.2 SDN Architecture

SDN has a **three-layer architecture**:

5.2.1 Layer 1 – Infrastructure Layer (Data Plane)

- Contains the physical network devices: switches, routers, access points.
- These devices are “**dumb**” **forwarders**: they simply forward packets according to flow rules installed by the controller.
- No distributed routing protocols (no OSPF, no STP running in the device).
- Each device has a **flow table** mapping packet header fields to actions (forward, drop, modify, send to controller).

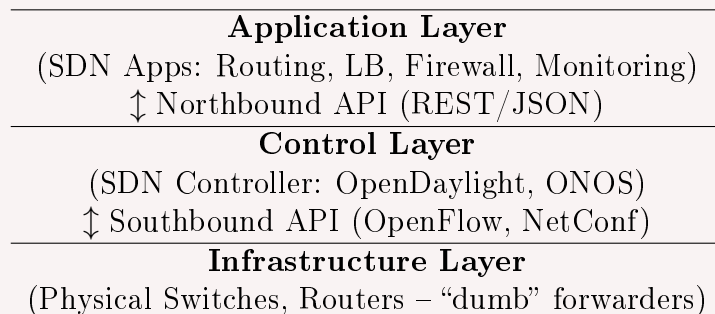
5.2.2 Layer 2 – Control Layer (Control Plane)

- Contains the **SDN Controller** (logically centralized, physically may be distributed for scalability).
- Has a **global view** of the entire network topology.
- Computes optimal paths and installs forwarding rules in devices via the southbound API.
- **Southbound API (Southbound Interface)**: Communication between controller and infrastructure layer. Examples: **OpenFlow**, NetConf, OVSDB.
- Popular SDN controllers: **OpenDaylight**, **ONOS**, Floodlight, Ryu, POX.

5.2.3 Layer 3 – Application Layer (Management Plane)

- Contains SDN applications that implement network services: routing, load balancing, firewalling, traffic engineering, monitoring.
- Applications communicate with the controller via the **northbound API**.
- **Northbound API (Northbound Interface)**: Typically RESTful HTTP/JSON API; allows applications to request network behavior.

SDN Architecture Diagram (Text)



5.3 OpenFlow Protocol

Definition: OpenFlow

OpenFlow (Open Networking Foundation – ONF) is the most widely used **south-bound protocol** in SDN. It defines a standard interface for the SDN controller to communicate with network devices, installing and managing **flow tables** in switches.

OpenFlow flow table entry:

[<u>Match Fields</u> <u>Priority</u> <u>Counters</u> <u>Actions</u> <u>Timeouts</u> <u>Cookie</u>]
Headers to match Rule precedence Packets, bytes Forward, drop, modify Hard/Idle Controller ID

OpenFlow match fields (examples): Ingress port, Ethernet src/dst, VLAN ID, IP src/dst, IP protocol, TCP/UDP port numbers, MPLS label, etc.

OpenFlow actions:

- **Output:** Forward packet to a specific port (or ALL, FLOOD, CONTROLLER).
- **Drop:** Discard the packet.
- **Modify-Field:** Rewrite header fields (NAT, VLAN tagging).
- **Send to Controller:** Punt first packet of new flow to controller for decision (*reactive forwarding*).

Reactive vs. Proactive flow installation:

- **Reactive:** First packet of unknown flow sent to controller; controller installs rules; subsequent packets forwarded by switch. High flexibility, but first-packet latency.
- **Proactive:** Controller pre-installs flow rules before any traffic arrives. Low latency, less flexible.

5.4 SDN Benefits

- **Centralized control:** Single logical view of the network; simplified management and troubleshooting.
- **Programmability:** Network behavior can be programmed via software (automation, CI/CD for networks).
- **Vendor independence:** Uses open southbound APIs; no vendor lock-in.

- **Dynamic traffic engineering:** Controller can re-route traffic in milliseconds based on real-time measurements.
- **Cost reduction:** Use commodity (white-box) switches instead of expensive proprietary hardware.
- **Security:** Centralized policy enforcement; rapid response to threats.
- **Innovation:** Network behavior can be changed by deploying new software – no hardware changes needed.

5.5 SDN Use Cases

- **Data center networking:** Google B4 (inter-datacenter WAN), Microsoft Azure, Facebook fabric all use SDN principles.
- **Network virtualization:** VMware NSX, Open vSwitch – create overlay networks for multi-tenant cloud environments.
- **WAN optimization:** Traffic engineering, load balancing across multiple links.
- **5G networks:** SDN enables flexible 5G core network slicing.
- **Campus/Enterprise networks:** Simplified VLAN management, security policy enforcement.

5.6 SDN vs. Traditional Networking

Table 5.1: Traditional Networking vs. SDN

Property	Traditional Networking	SDN
Control plane	Distributed (in each device)	Centralized (SDN Controller)
Data plane	In each device	In each device (unchanged)
Configuration	Per-device CLI / SNMP	Controller API (REST)
Intelligence	In the device (proprietary)	In the controller (software)
Protocols	Vendor-specific + standards	Open APIs (OpenFlow, REST)
Flexibility	Low (firmware/hardware changes)	High (software changes)
Network view	Local (per-device)	Global
Vendor lock-in	High	Low (white-box hardware)
Scalability	Horizontal (add more devices)	Controller scalability needed
Fault tolerance	Distributed (resilient)	Single point of failure risk (mitigated by controller clustering)

6 Interconnection: Cloud, IoT, and SDN

Key Concept: Convergence of Technologies

Cloud Computing, IoT, and SDN are deeply interconnected in modern networks:

- **IoT + Cloud:** IoT devices generate massive data; the cloud provides the processing power, storage, and analytics. Platforms like AWS IoT Core, Azure IoT Hub, and Google Cloud IoT use MQTT/CoAP to ingest data from billions of devices into cloud backends.
- **IoT + SDN:** SDN can manage the heterogeneous, large-scale IoT network infrastructure: dynamic QoS policies for critical IoT traffic (medical devices), isolating IoT device traffic for security, and auto-provisioning connectivity for new devices.
- **SDN + Cloud:** Cloud data centers use SDN for virtual network management (VPC routing, security groups). NFV (Network Functions Virtualization) uses SDN principles to run virtual firewalls, load balancers, and routers as software.
- **Edge Computing:** IoT data is increasingly processed at the **edge** (near the device) using fog/edge nodes – reducing latency and bandwidth to the cloud. SDN manages edge-to-cloud traffic orchestration.

7 Practice Questions (MAKAUT Style)

Short Answer Questions (2–3 marks)

- Q1.** What is cloud computing? State the five key characteristics defined by NIST.
- Q2.** Differentiate between IaaS, PaaS, and SaaS with two examples each.
- Q3.** Compare Public, Private, and Hybrid cloud deployment models.
- Q4.** What is IoT? List any four application domains of IoT.
- Q5.** What are the main challenges of IoT networking?
- Q6.** What is MQTT? Explain the publish-subscribe model.
- Q7.** Explain the three QoS levels in MQTT with their delivery guarantees.
- Q8.** What is the MQTT Last Will and Testament (LWT) feature?
- Q9.** What is CoAP? How does it differ from HTTP?
- Q10.** Explain the four CoAP message types: CON, NON, ACK, RST.
- Q11.** What is SDN? How does it differ from traditional networking?
- Q12.** Explain the three layers of SDN architecture.
- Q13.** What is OpenFlow? What are flow table entries?
- Q14.** Differentiate between MQTT and CoAP.

Long Answer Questions (5 marks)

- Q1.** Explain cloud computing with its service models (IaaS, PaaS, SaaS) and deployment models. What is a VPC? How does virtualization support cloud networking?
- Q2.** Explain the IoT architecture in detail (4-layer model). Describe the key wireless technologies used in IoT and compare them.
- Q3.** Explain MQTT protocol in detail: architecture, publish-subscribe model, QoS levels, packet types, LWT, retained messages, and security mechanisms.
- Q4.** Explain CoAP protocol: design principles, message types, request-response model, Observe extension, and security (DTLS). Compare MQTT and CoAP.
- Q5.** Explain Software-Defined Networking (SDN): motivation, three-layer architecture, OpenFlow protocol, and benefits. Compare SDN with traditional networking.

8 Module 7 – Key Takeaways

Summary Checklist – Module 7

- ✓ Cloud: on-demand, elastic, pay-per-use; 5 NIST characteristics.
- ✓ IaaS (VMs) $\subset$ PaaS (Runtime) $\subset$ SaaS (Full app).
- ✓ Public (shared) / Private (dedicated) / Hybrid (mix) / Community.
- ✓ Cloud networking: VPC, vSwitch, overlay (VXLAN), load balancer, CDN.
- ✓ IoT: 4 layers – Perception / Network / Processing / Application.
- ✓ IoT challenges: scale, heterogeneity, security, power, addressing.
- ✓ MQTT: publish-subscribe; broker; topics (wildcards + and #); TCP/1883.
- ✓ MQTT QoS: 0 (fire-forget), 1 (at-least-once), 2 (exactly-once, 4-step).
- ✓ MQTT: LWT (detect ungraceful disconnect), retained messages, persistent sessions.
- ✓ CoAP: RESTful; UDP/5683; 4-byte header; CON/NON/ACK/RST types.
- ✓ CoAP: GET/POST/PUT/DELETE; Observe (subscribe); DTLS for security.
- ✓ MQTT vs CoAP: broker vs broker-less; TCP vs UDP; push vs pull.
- ✓ SDN: decouples control plane (controller) from data plane (forwarder).
- ✓ SDN layers: Infrastructure (data) / Control (OpenDaylight, ONOS) / Application.
- ✓ OpenFlow: southbound API; flow table = match + priority + actions.
- ✓ SDN benefits: centralized view, programmable, vendor-independent, dynamic.